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[Intervention Review]

Interventions for smoking cessation in inpatient psychiatry settings

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ABSTRACT

Rationale

Tobacco smoking is a leading preventable cause of premature morbidity and mortality in people living with severe mental illness. Smoking rates are disproportionately high and abstinence rates low in people with severe mental illness, despite their reported interest in quitting being comparable to that of the general population. Inpatient psychiatry settings have been identified as opportune places to promote and support tobacco cessation amongst people with severe mental illness who smoke, particularly since many hospitals have implemented smoke-free policies. Evidence is needed to guide policy and practice for tobacco cessation interventions within inpatient psychiatry settings.

Objectives

To assess the effects of smoking cessation interventions on tobacco smoking in adults receiving inpatient psychiatry treatment.

Search methods

We searched the following bibliographical databases and clinical trial registers from inception until 10 February 2026: Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE, Embase (Elsevier), PubMed, PsycINFO (EBSCOhost), CINAHL Complete (EBSCOhost), ProQuest Dissertations and Theses Global, ClinicalTrials.gov, World Health Organization International Clinical Trials Registry Platform (WHO ICTRP). We also handsearched the annual meeting abstracts for the Society for Research on Nicotine and Tobacco (SRNT) and screened reference lists of eligible studies.

Eligibility criteria

We included randomised controlled trials (RCTs) and cluster-RCTs that assessed interventions for tobacco cessation amongst people of 18 years and older who were inpatients in psychiatry settings. Interventions had to be initiated in the psychiatry inpatient setting and aimed at supporting smoking cessation.

Outcomes

Our critical outcome was smoking abstinence at six months (biochemically verified) and our important outcomes included serious adverse events.

Risk of bias

We used the Cochrane risk of bias tool (RoB 2) to assess the outcomes of our review.

Synthesis methods

We synthesised results using meta-analysis where appropriate, calculating risk ratios using the inverse-variance random-effects model. Where this was not possible, we synthesised results following Synthesis Without Meta-analysis (SWiM) guidelines. We used GRADE to assess our level of certainty in the evidence for our two key outcomes.

Included studies

We included 10 studies that involved 2262 people in total. Three trials were conducted in the USA, two in Australia, two in Taiwan, two in Iran, and one in Israel. The studies took place in emergency or acute and long-stay psychiatric settings. In most studies, the participants had a mix of diagnoses (e.g. mood disorders, anxiety disorders, schizophrenia), and three studies involved only participants with schizophrenia or schizophrenia-type disorders.

Five studies tested smoking cessation counselling plus nicotine replacement therapy with post-discharge follow-up support versus usual care; one study tested a group behavioural programme for smoking reduction versus waitlist control; and four studies tested pharmacotherapy interventions including smoking medications (bupropion versus placebo, cytisine versus nicotine replacement therapy) and nicotine replacement therapy (different types and doses).

Synthesis of results

We found low-certainty evidence of increased smoking abstinence from interventions that provided counselling with nicotine replacement therapy and post-discharge support, compared to usual care, when measured six months after the start of the intervention or hospital discharge (RR 1.81, 95% CI 1.33 to 2.47; $P < 0.001$, $I^2 = 0\%$; 5 studies, 1611 participants; low-certainty evidence).

Across studies, serious adverse events (SAEs) were low. We pooled four studies reporting deaths at 6 to 18 months after the start of the intervention or hospital discharge. We found that the intervention of smoking cessation counselling plus nicotine replacement therapy with post-discharge support may result in a slight reduction in deaths compared to usual care, but the results are very uncertain due to low event numbers (22), considerable imprecision, and some concerns about risk of bias (RR 0.81, 95% CI 0.34 to 1.96; $P = 0.50$, $I^2 = 0\%$; 4 studies, 1431 participants; very low certainty evidence).

Authors' conclusions

People receiving inpatient psychiatry treatment may be more likely to have successfully stopped smoking six months after the inpatient intervention when offered smoking cessation counselling plus nicotine replacement therapy with continued post-discharge support, compared with usual care, but the certainty of the evidence is low.

There was insufficient evidence to determine the effectiveness of other smoking cessation interventions initiated in the psychiatry inpatient setting. More randomised controlled trials, especially those evaluating pharmacological interventions, are needed to strengthen conclusions about treatment effects.

Funding

This Cochrane review had no dedicated funding.

Registration

Protocol (2024) DOI: 10.1002/14651858.CD015934

PLAIN LANGUAGE SUMMARY

Do programmes for quitting smoking (smoking cessation) work for people who are in hospital for treatment of mental illness?

Key messages

- Offering counselling plus nicotine replacement therapy (a treatment that provides nicotine in a safer form than tobacco smoking to help people reduce cravings and withdrawal symptoms), during a stay in a psychiatric ward, and continuing this support after leaving hospital, may help more people to stop (quit) smoking tobacco (when measured six months later) than usual care.
- We did not find enough information to draw conclusions about the effects of any other smoking cessation interventions initiated in the psychiatry inpatient setting. We need more, better designed studies that test different ways to help people quit smoking, particularly medicines.

Why is smoking a problem for people with mental illness?

People experiencing mental illness die up to 15 to 20 years earlier than people in the general population from preventable physical illnesses, such as heart and blood vessel disease, airway and lung disease, diabetes, and cancer. Smoking tobacco is one of the main causes of these illnesses. People being treated in hospital for mental illness smoke more than people in the general population and are less successful at quitting, despite wanting to stop. Many hospitals are smoke-free environments and should be a good place to deliver quit-smoking programmes to people with mental illness.

What did we want to find out?

We wanted to know whether quit-smoking programmes started while people are being treated in a hospital psychiatry ward help these people to stop smoking, and which programmes are most effective. We were particularly interested in how many people were still not smoking after six months and whether anyone suffered any serious unwanted effects, such as death or a life-threatening event.

What did we do?

We searched for studies that investigated any quit-smoking programme started in a psychiatric ward compared to usual care, being put on a waiting list, or receiving a placebo (a 'fake' medicine with no active ingredients). People taking part could have any psychiatric diagnosis but had to be adults (aged 18 years and older) who were tobacco smokers.

We summarised the results of the studies and rated our confidence in the evidence, based on factors such as study methods and sizes.

What did we find?

We found 10 studies that included a total of 2262 people in emergency or long-stay psychiatry wards. Most studies included people with a variety of mental health diagnoses (for example, mood disorders, anxiety disorders, schizophrenia), and three studies involved only people with schizophrenia or schizophrenia-type disorders. The studies took place in five countries (Australia, USA, Taiwan, Israel, and Iran).

The programmes included:

- counselling plus nicotine replacement therapy and continued support with quitting smoking after leaving hospital (versus usual care);
- a group behavioural (psychological support without medicine) smoking reduction programme (versus being put on a waiting list);
- medicines including bupropion (versus placebo) or cytisine (versus nicotine replacement therapy);
- comparisons between different types and doses of nicotine replacement therapy.

Main results

Counselling plus nicotine replacement therapy together with continued support after leaving hospital compared with usual care (5 studies, 1611 people):

- may increase the number of people who are still not smoking six months after leaving hospital by about 6 more people per 100; and
- may result in fewer deaths (1 less death per 100 people), but this result is very uncertain.

What are the limitations of the evidence?

Our results are based on a limited number of studies, and our confidence in the evidence is low. We had very low confidence in the result for deaths because there were very few deaths in the studies.

Anti-smoking medicines, such as varenicline, have previously been shown to work for people with mental illness in the community, but we found no studies that tested their use in a hospital psychiatry ward.

How up to date is this evidence?

This review is up to date to 10 February 2026.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Smoking cessation counselling plus nicotine replacement therapy with post-discharge support vs. Usual care for adult psychiatry inpatients

Patient or population: adult psychiatry inpatients

Setting: Australia and USA

Intervention: Smoking cessation counselling plus nicotine replacement therapy with post-discharge support

Comparison: Usual care

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with Usual care	Risk with Smoking cessation counselling plus nicotine replacement therapy with post-discharge support				
Smoking abstinence at 6 months	7 per 100	13 per 100 (9 to 17)	RR 1.81 (1.33 to 2.47)	1611 (5 RCTs)	⊕⊕⊕⊕ Low ^a	Smoking cessation counselling plus nicotine replacement therapy with post-discharge support may increase abstinence at 6 months (about 6 more people per 100) compared to usual care.
Serious adverse events (deaths) follow-up: range 6 months to 18 months	2 per 100	1 per 100 (1 to 4)	RR 0.81 (0.34 to 1.96)	1431 (4 RCTs)	⊕⊕⊕⊕ Very low ^{b,c}	Smoking cessation counselling plus nicotine replacement therapy with post-discharge support may result in a slight reduction in Serious Adverse Events (deaths), but the evidence is very uncertain.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_460075647693480262.

- ^a Downgraded two levels for very serious risk of bias: three of five contributing studies were judged at high risk of bias due to missing outcome data. The remaining studies had some concerns, substantially reducing confidence in the effect estimate.
- ^b Downgraded two levels for very serious imprecision: few events and wide confidence interval crossing no effect, encompassing important benefit and harm.
- ^c Downgraded one level for some concerns of risk of bias in three of four studies.

BACKGROUND

Tobacco smoking is a leading cause of preventable early mortality and morbidity for people experiencing mental illness [1], contributing to cardiometabolic disease burden and shortening life expectancy by 10 to 15 years [2, 3]. Smoking compounds other risk factors for disease that are prevalent in this population, such as obesity [4], and contributes to social disadvantage and financial hardship for individuals [5, 6]. Smoke-free psychiatry inpatient settings are well-situated to initiate evidence-based treatment for tobacco withdrawal management, and to support ongoing abstinence from smoking. Reducing the physical health inequities experienced by people with mental illness requires clinical practice guidelines that support the delivery of setting-appropriate smoking cessation interventions.

Description of the condition

Tobacco smoking rates for people experiencing mental illness are two to three times higher than those of the general population [7, 8, 9]. This disparity is demonstrated internationally [10, 11, 12], with some of the highest rates seen amongst people receiving treatment in psychiatry inpatient settings [13, 14, 15]. People with mental illness face unique challenges when it comes to stopping smoking. They are more likely than the general population to smoke more heavily, have greater nicotine dependence, and experience multiple failed quit attempts and less cessation success [7, 8, 11, 16, 17, 18]. Tobacco-related diseases, such as cardiovascular disease, chronic obstructive pulmonary disease, lung cancer, and stroke, result in a five-fold increase in mortality risk for people with mental illness [2, 19], which makes it a priority to help this population reduce tobacco use [20, 21].

Description of the intervention and how it might work

Tobacco smoking cessation interventions initiated in hospital and continued for over one month have been shown to increase quit rates for hospital patients in general wards who smoke [22]. Rigotti and colleagues developed a hierarchy of hospital smoking cessation interventions based on the intensity and continuity of support provided during admission and after discharge, recognising post-discharge follow-up as a key component of effective treatment [23]. Combined behavioural and pharmacological intervention for smoking cessation is recommended in clinical practice guidelines for both general patients and those receiving psychiatry treatment [24, 25]. Pharmacotherapy interventions include nicotine replacement therapies (NRT) and non-nicotine medicines such as varenicline, bupropion, and cytisine, with evidence of bupropion and varenicline potentially offering improved cessation outcomes for people with mental illness [26, 27]. Behavioural interventions include brief advice, self-help resources, or counselling (e.g. motivational interviewing). These interventions can be delivered individually or in groups and may consist of a single session, multiple sessions, or a combination of components. Recent evidence has indicated that alternative products that deliver nicotine, such as electronic cigarettes [28], may increase cessation of tobacco smoking, including for people with mental illness [29, 30, 31]; however, the evidence base is limited.

Smoke-free psychiatry inpatient units provide the opportunity for a period of unplanned abstinence from smoking and for smoking cessation treatment under medical supervision. Hospital-delivered

tobacco treatment interventions typically follow a 5-As approach (Ask, Advise, Assess, Assist, and Arrange) [28], which is also applied in psychiatry settings [32]. Most acute psychiatric admissions are brief, with people rarely planning to quit smoking at the time of admission, and tobacco treatment focused on managing nicotine withdrawal [33, 34], with little to no referral for continued cessation support after hospital discharge [14]. Since post-discharge relapse rates are high [35, 36], conventional approaches may be insufficient to ensure cessation for people with mental illness who smoke [11].

Why it is important to do this review

Historically, smoking has been accepted in psychiatry settings [5, 33, 37], which has meant that smoking has not routinely been addressed in healthcare, despite evidence that smoking cessation treatments are safe, effective and can improve physical and mental health [27, 38, 39, 40]. Factors identified as contributing to limited provision of smoking cessation interventions in routine care include competing clinical priorities, limited resources, and a lack of staff training in smoking cessation interventions [41]. In addition, features of the psychiatric-inpatient environment, including restricted autonomy, symptom acuity and involuntary admission, can influence how cessation interventions are delivered and used and how effectively they work in practice. People with mental illness are motivated to stop smoking ('quit') and can engage in cessation interventions during psychiatric hospitalisation [11, 33, 34], but few will ask for support or be offered interventions [14]. Many reviews of the evidence on smoking cessation interventions in hospital settings have generally excluded people with mental illness [22], while others have focused on specific interventions [26, 42, 43], population subgroups [44, 45, 46, 47], outpatient settings [35, 48]. Meta-analysis is not always included [49]. This Cochrane review specifically addresses smoking cessation interventions delivered in the inpatient psychiatry setting and provides an up-to-date synthesis of the evidence that is inclusive of all psychiatric diagnoses, smoking cessation interventions and emerging interventions. To optimise the delivery of smoking cessation for people with mental illness and to support clinicians delivering smoking cessation care, it is necessary to understand the smoking cessation interventions that work best in this setting.

OBJECTIVES

To assess the effects of smoking cessation interventions on tobacco smoking in adults receiving inpatient psychiatry treatment.

To assess whether the effects of smoking cessation interventions differ according to psychiatric diagnosis or type of intervention or comparator intervention.

METHODS

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) when conducting the review and PRISMA 2020 for the reporting [50, 51].

Differences between protocol and review

We made some minor departures from protocol, which are justified below.

1. Search strategy modification: the review protocol specified that we would search PsycEXTRA (EBSCO), but this database was not

accessible at the time of the review. To maintain coverage of psychological and behavioural literature, we searched CINAHL Complete and APA PsycINFO (both via EBSCOhost) using the prespecified strategy.

2. Risk of bias (RoB) assessors: although the review protocol specified that there would be two RoB assessors, additional review authors contributed to RoB assessments to ensure timely completion; all assessments were conducted independently and in duplicate.
3. Subgroup analyses: we did not conduct planned subgroup analyses due to insufficient data and substantial heterogeneity. Sex/gender-based subgrouping was not possible because studies lacked sex/gender-stratified outcomes or evidence of sex/gender as an effect modifier. No included trials enrolled participants under 18 years of age.
4. Outcome definition adjustments
 - a. We used point-prevalence abstinence in preference to sustained cessation rates for abstinence outcomes because only two studies reported sustained abstinence and results were reported narratively.
 - b. We amended the prespecified end-of-intervention abstinence time point due to overlap with six-month follow-up data in two trials. Instead we used the earliest post-intervention follow-up to avoid double counting.
 - c. In the protocol, we defined serious adverse events as a composite outcome that included deaths and rehospitalisation. In the review, we meta-analysed only deaths, as this outcome was reported consistently across studies, and we used the final point of data collection to provide the most comprehensive data. Sensitivity analyses based on the originally prespecified end-of-intervention time points showed no important differences from the primary analysis, confirming that the amendment for follow-up data did not impact the review's conclusions. Rehospitalisation was narratively reported.
 - d. We extended the timeframe stated in the protocol for adverse events as several studies did not provide six-month outcomes.
5. Summary of findings table 1: we changed the time point for serious adverse events from 'at six months', as specified in the protocol, to 'six to 18 months'.
6. Summary of findings tables: we have not provided summary of findings tables for the following comparisons: pharmacotherapy versus control and behavioural interventions versus control.

Criteria for considering studies for this review

Types of studies

We searched for randomised controlled trials (RCTs), including cluster-RCTs and cross-over RCTs, that compared smoking cessation interventions with placebo, usual/standard care, a different intervention, waitlist control, or no-intervention control. We did not include quasi-randomised studies. We also included trials that compared different regimens of an intervention, more than one type of intervention versus a single type, or combinations of these. We included data from all types of publications without any language or date restrictions.

Types of participants

Participants were adults (aged 18 years and older), with any psychiatric diagnosis, who smoked tobacco and were inpatients in a psychiatry unit and provided consent to participate. Our protocol stated we would include trials recruiting adults that included participants under 18 years of age provided the mean age of participants was at least 18 years. However, no trials involving participants under 18 years were identified. We excluded trials recruiting participants with substance use disorders (other than nicotine addiction), learning disability, dementia, other neurocognitive disorders, or terminal illness without any other mental illness. We excluded trials recruiting from psychiatry outpatient settings and participants who exclusively used e-cigarettes.

Types of interventions

We included any tobacco cessation intervention that was initiated during inpatient admission and delivered by hospital and/or research staff. Eligible interventions included behavioural therapy (e.g. brief interventions, multi-session individual and group counselling); smoking cessation pharmacotherapy including nicotine replacement therapy (NRT) of any modality (e.g. gum, patches, inhalers, lozenge); varenicline, bupropion, nortriptyline, cytosine; and any combinations of these interventions. Our search included e-cigarette interventions to reduce tobacco use, but none were identified for inclusion in this review.

We included trials that compared tobacco cessation interventions with any comparator, as described above. We did not include trials that considered smoking bans in psychiatry settings as an intervention for smoking cessation.

We used the following comparison groups.

- Intervention versus control (no treatment, treatment as usual/standard care, minimal intervention, delayed/waitlist control)
- Intervention versus placebo
- Intervention A versus intervention B

Outcome measures

We included trials of the above-mentioned interventions that measured our critical or important outcomes. We contacted study authors to ascertain availability of unpublished trial outcomes for inclusion in the meta-analyses.

Critical outcomes

The critical outcome of interest was smoking abstinence at six months from the start of the intervention or hospital discharge. In our protocol, we had planned to measure sustained cessation rates in preference to point-prevalence abstinence. However, this outcome was reported in only two studies and definitions varied. Therefore, we used point-prevalence abstinence at common time points, enabling meta-analysis. We summarised sustained abstinence results narratively, where available. We preferred biochemically validated to self-reported abstinence. These were considered on an intention-to-treat (ITT) basis, with participants lost to follow-up treated as continuing smoking.

Important outcomes

As reported by Stockings 2019 [52], short-term abstinence may indicate meaningful change in smoking for people with mental illness, who have lower quit rates than the general population [53]. Consequently, we also examined the following outcomes where reported.

- Abstinence at the end of the intervention (where the intervention assessment is not the six-month assessment). We used biochemically validated abstinence (where reported) in preference to clinician-scored measures, in preference to self-report measures.
- Proportion of participants taking up smoking cessation interventions (e.g. Quitline) at the end of the intervention
- Change in the number of cigarettes consumed per day at the end of the intervention
- Validated quality of life (QoL) measures assessing mental health, social quality of life, isolation, or loneliness at the end of the intervention were included. We prioritised smoking-specific measures (e.g. Tobacco Quality of Life Impact Tool), followed by generic or disease-specific measures commonly used in mental health populations, such as the WHOQOL-BREF and 36-item Short Form Health Survey (SF-36), and then the EQ-5D or other validated QoL instruments [54].
- Adverse events (e.g. aggression/violent incidents as recorded by staff at the study sites, mental illness symptoms, weight gain, or adverse effects of any smoking cessation medicines administered as part of the intervention) assessed as the number of participants having experienced an adverse event by six months. We extended the timeframe stated in the protocol to the longest follow-up as several studies did not provide six-month outcomes.
- Serious adverse events (SAE): in our protocol, we planned to include any event that led to death, was life-threatening, required inpatient hospitalisation or prolongation of existing hospitalisation, or resulted in persistent significant disability. We found few studies reporting rehospitalisation and definitions varied. To enable synthesis, we reported deaths and rehospitalisation separately, applying meta-analysis and narrative analysis as appropriate, and included events up to the longest follow-up.

Search methods for identification of studies

Electronic searches

As specified in our protocol [55], we searched the relevant databases for trials of smoking cessation interventions for psychiatry inpatients, identifying studies with a psychiatry diagnosis in the title, abstract, or keywords. We ran our search strategy on 14 October 2024 and again on 10 February 2026.

The search strategy combined MeSH and indexing terms and free-text words across the title, abstract, or keyword fields. Search terms were grouped around research design (RCTs), psychiatric illness, intervention (i.e. tobacco, smoking cessation), and psychiatry hospital settings. Search strategies for this review are listed in [Supplementary material 1](#). We included published and unpublished research studies. We did not limit our searches by language, year of publication, or publication format. We used Google Translate to translate studies not published in the English language.

- Cochrane Central Register of Controlled Trials (CENTRAL; 2024, Issue 10) in the Cochrane Library includes the Cochrane Tobacco Addiction Review Group Specialised Register (latest issue to 10 February 2026)
- MEDLINE (Ovid 1946 to present)
- Embase.com (Elsevier, 1947 to present)
- PubMed (MEDLINE coverage includes 1966 to present)
- APA PsycINFO (EBSCOhost, 1600 to present)
- CINAHL Complete (Cumulative Index to Nursing and Allied Health Literature) (EBSCOhost, 1937 to present)
- Dissertations and Theses Global (ProQuest 1637 to present)

PsycEXTRA was not accessible at the time of the search, but we undertook comprehensive searches of CINAHL Complete and APA PsycINFO (both via EBSCOhost) using the prespecified strategy.

Searching other resources

We searched international online clinical trials registers for ongoing and recently completed trials, including ClinicalTrials.gov (<https://clinicaltrials.gov>) and the World Health Organization International Clinical Trials Registry Platform (WHO ICTRP) (<https://trialssearch.who.int/>) to 14 October 2024 and again on 10 February 2026.

We manually searched annual meeting abstracts for the Society for Research on Nicotine and Tobacco (SRNT) (2020 to 2025) to identify early reports of ongoing research in the field using search terms including smoking, tobacco, cessation, mental illness, inpatient, psychiatric, hospitalisation, RCT.

We have listed trial registers or protocols for in-progress trials eligible for inclusion in the review in a 'Characteristics of ongoing studies' table ([Supplementary material 4](#)). When we conducted the top-up search in February 2026, we contacted authors of ongoing studies to see if study data were available.

We searched relevant grey literature, including sources such as reports, dissertations, theses, and conference abstracts using ProQuest Dissertations and Theses Global (PQDT). We searched within existing reviews and any relevant systematic reviews on the same topic and checked reference lists of included studies.

We undertook post-publication checks to identify any amendments, retractions, expressions of concern, or other indicators of problematic studies. One included study, Brown 2021 [56, 57], issued amendments in June 2021 and November 2021 to correct statistical results and table errors; we used the corrected version in all analyses. No other issues were identified.

Data collection and analysis

All study data, analyses, and risk-of-bias tables entered in Review Manager (RevMan) [58] are available as the Cochrane RevMan data package ([Supplementary material 7](#)).

Selection of studies

Three authors (JYT, PT, SP), in pairs, independently screened titles and abstracts of retrieved studies (de-duplicated) for relevance for inclusion using Covidence software [59]. The same authors then retrieved and independently reviewed in duplicate the full text of all studies not excluded during title and abstract screening. Any disagreements were resolved by discussion or by consulting a third

review author (CG). We reported the selection process in a PRISMA diagram ([Figure 1](#)).

Figure 1. PRISMA study flow diagram

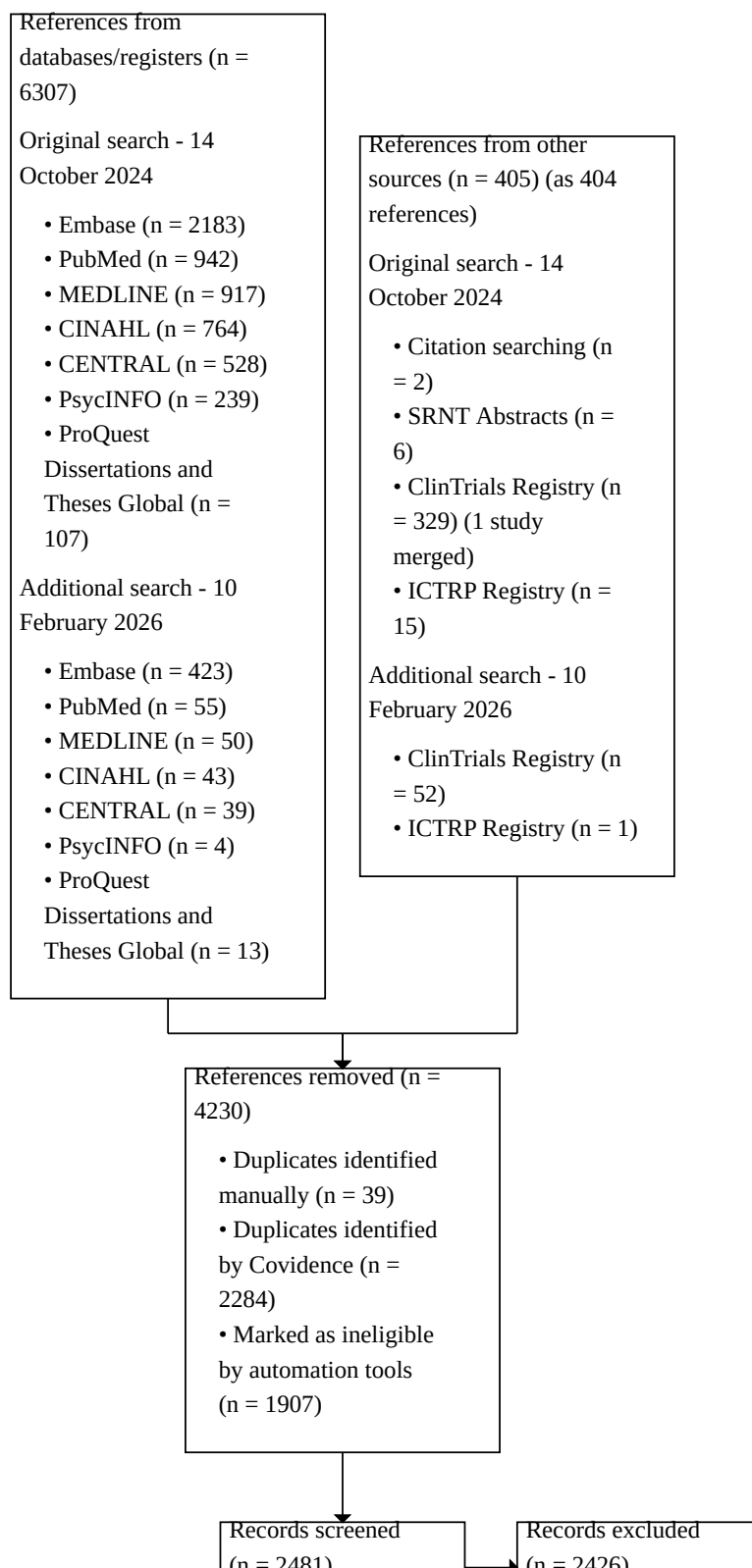
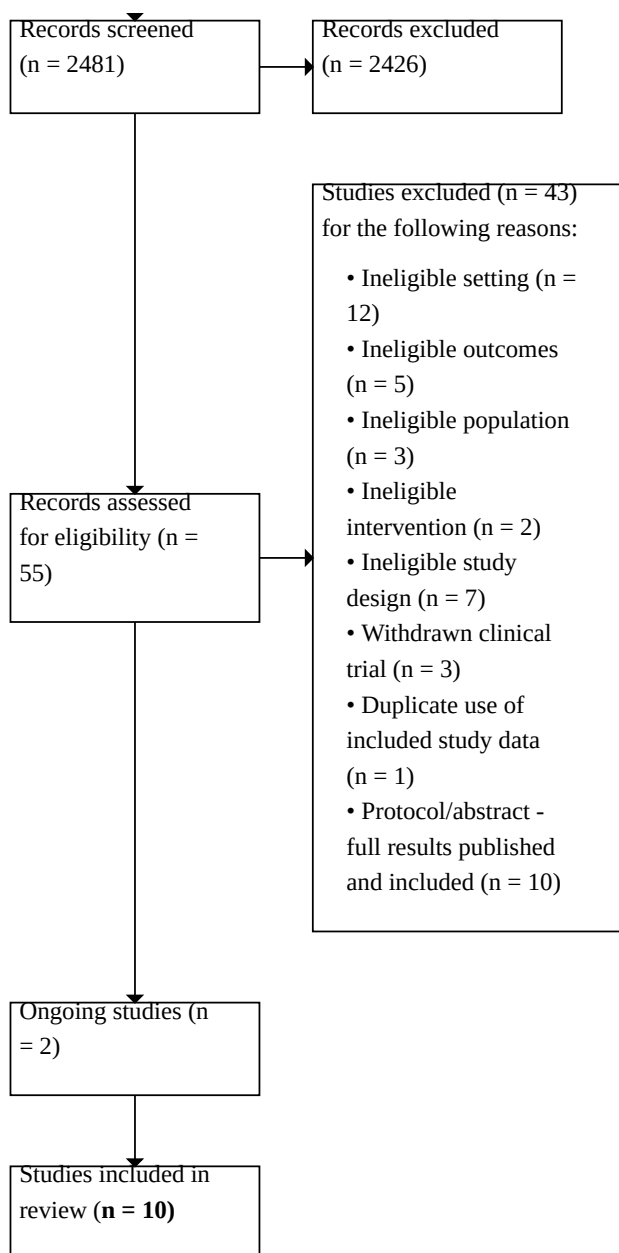


Figure 1. (Continued)



We categorised reports that did not meet inclusion criteria following full-text screening as excluded studies and listed them in a 'Characteristics of excluded studies' table ([Supplementary material 3](#)), along with the reasons for their exclusion.

Data extraction and management

Two review authors (SP, AG) independently extracted data from the included studies using a data extraction sheet in Microsoft Excel [60]. We resolved any disagreements by discussion or by consulting a third review author (CG). For each study, we extracted the following information, where available, and summarised this in our analysis and in a 'Characteristics of included studies' table ([Supplementary material 2](#)).

- Study details: authors' names, year of publication, country
- Methods: study design, recruitment period and procedure, recruitment setting(s), number of study locations if multiple, study start and end date
- Participants: number (N) (intervention/control), N consented, N analysed, N admission status (voluntary/involuntary), inclusion and exclusion criteria, randomisation, definition of smoker used, psychiatric diagnosis, demographic characteristics (e.g. mean or median age, age range, sex/gender, ethnicity), mean cigarettes per day, mean Fagerström Test for Nicotine Dependence (FTND) [61] score or other measure of nicotine dependence (e.g. Heaviness of Smoking Index (HSI)) [62], motivation to quit at study intake (as judged by eligibility

criteria, author report, scores on motivation measures, and type of participant enrolment)

- Interventions: description of intervention(s) (treatment type, dosage, number and duration of sessions, designation of intensity as defined by Rigotti and colleagues [23], follow-up support), description of control (treatment, dosage, regimen, behavioural support), what comparisons were carried out between which groups, concomitant medications, and excluded medications
- Outcomes: primary and secondary outcomes specified and collected, type of biochemical validation (if any), definitions of abstinence and smoking reduction, proportion of participants with follow-up data, follow-up length, timing of follow-up assessments, ITT and per-protocol analysis data, adverse events and serious adverse events
- Risk of bias: see [Risk of bias assessment in included studies](#)
- Notes: trial funding and any trial author declared or potential conflicts of interest and additional comments

Risk of bias assessment in included studies

Four review authors (SP, JYT, PT, DY) independently, and in pairs, assessed the risk of bias (RoB) in the included studies for the outcomes specified in our review, using the RoB 2 tool, as recommended in Chapter 8 of the *Cochrane Handbook for Systematic Reviews of Interventions* [63].

We conducted risk of bias assessments for our critical outcome (smoking abstinence at six months) and for important outcomes that contributed to our review's evidence synthesis (smoking abstinence at end of intervention, cigarettes smoked per day, and serious adverse events). We did not assess other important outcomes using RoB 2 because they were too clinically and methodologically heterogeneous to be synthesised in any form and therefore did not contribute effect estimates to the review. Our review did not have any included studies with a cross-over design, and the one included study that was cluster-randomised did not contribute any analysed outcomes.

In line with the `outcome-specific` structure of RoB 2, we made separate judgements for each study-outcome pair. We evaluated bias for the effect of assignment to intervention (`intention-to-treat` effect), consistent with the review's focus on estimating the effect of being allocated to the intervention rather than the effect of adherence.

We used Cochrane RoB 2 domains [64], as follows.

- Bias arising from the randomisation process
- Bias due to deviations from intended interventions
- Bias due to missing outcome data
- Bias in measurement of the outcome
- Bias in selection of the reported result

We used the RoB 2 Template and Excel tool to complete and manage assessments [65], with each review author (SP, JYT, PT, DY) independently assessing each domain using the RoB 2 series of signalling questions and answers (yes, probably yes, no information, probably no, no) [66]. Risk of bias was determined by responses to signalling questions and RoB 2 algorithms, and we accompanied our decision justifications with direct quotes or study details in a risk of bias table, recording domain-level judgements

of low risk of bias, some concerns, or high risk of bias. As per the *Cochrane Handbook for Systematic Reviews of Interventions* [67], studies were low risk of bias overall if they were low risk for all domains; high risk of bias if judged at high risk in one or more domains or had some concerns in multiple domains; and some concerns if not at high risk in any domain but some concerns in at least one domain. The overall risk of bias judgement for review outcomes was based on consensus, with disagreements resolved by discussion or a third review author (CG). Summary results for each included review outcome and associated study are displayed in a risk of bias 'traffic light' with each analysis.

Measures of treatment effect

We pooled treatment effects for dichotomous outcomes (e.g. smoking abstinence) and reported risk ratios (RRs) using the following formula: (number of events in the intervention condition/ intervention denominator)/(number of events in control condition/ control denominator) with 95% confidence intervals (CIs) [65]. No data transformations or additional calculations were required for the meta-analysed dichotomous outcomes. Where the event is defined as smoking cessation, an RR greater than one indicates more abstinent people in the treatment group than in the control group.

We did not meta-analyse continuous outcomes; we summarised them narratively using the summary statistics reported in the original studies. For one continuous outcome, we converted a reported standard error to a standard deviation to allow visual comparison with results from other studies in a graphical presentation. This conversion was undertaken for display purposes only and was not used in any meta-analysis. When studies did not report effect estimates directly, we extracted available summary statistics and results were narratively summarised.

Unit of analysis issues

The unit of analysis was the individual participant. Had the cluster-randomised trial we identified been included in any analysis, we would have made appropriate adjustments. In studies where results were presented for multiple follow-up periods, we noted the follow-up period that we selected.

Dealing with missing data

By convention, we considered participants who dropped out or were lost to follow-up as having continued to smoke. We excluded any deaths from the total sample size for analyses where appropriate. We noted adverse events in the [Results](#) section of the review.

Reporting bias assessment

It was not appropriate to create a funnel plot to assess potential publication bias or small-study effects as none of our analyses pooled 10 or more studies [68]. We searched for and reported studies that we knew were completed but for which results were unavailable.

Synthesis methods

We considered clinical and methodological heterogeneity by inspecting all studies for clearly outlying study characteristics, participant characteristics, intervention type, methods, or outcomes. We grouped studies by intervention type and conducted

meta-analysis when a group of studies was considered sufficiently homogeneous to provide a meaningful summary. We grouped the studies into the following comparisons.

1. Counselling plus pharmacotherapy (NRT) with post-discharge support versus usual care
2. Group behavioural smoking reduction programme versus waitlist
3. Pharmacotherapy: smoking medication versus placebo/NRT
4. Pharmacotherapy: nicotine replacement therapy (different forms or doses)

We used inverse variance random-effects models for all analyses. We used the I^2 statistic in RevMan [58] to assess statistical heterogeneity, given by the formula $[(Q - df)/Q] \times 100\%$, where Q is the χ^2 statistic and df is the associated degrees of freedom, and described heterogeneity in the pooled estimates as being low, moderate, or high according to the values 25%, 50%, and 75%, respectively [69]. We considered the risk of bias in our interpretation and GRADE certainty assessment.

For studies with outcomes where meta-analysis was not considered possible, due to clinical or methodological heterogeneity (or both), we used the Synthesis Without Meta-analysis (SWiM) guidelines to provide a consistent narrative synthesis of results [70]. We used structured tabulation for each review outcome for each study's reported data (measure, time point, direction), ordering by outcome and study. This was accompanied by a brief narrative interpretation. We incorporated risk of bias assessments into the interpretation for those outcomes that underwent Rob 2 evaluation. Outcomes not assessed with Rob 2 were narratively summarised without risk-of-bias weightings.

Investigation of heterogeneity and subgroup analysis

We considered both clinical and methodological heterogeneity, and studies were discussed by the review author team regarding whether they should be included in meta-analysis (SP, SK, CG). We did not conduct subgroup analysis by sex/gender because the studies contributing to this comparison did not provide sex/gender-stratified outcome data or indicate that sex/gender acted as an effect modifier for smoking cessation outcomes. None of the included studies involved participants under 18 years of age.

Equity-related assessment

We did not investigate health inequities in this review (and had not planned to do so in our protocol). The included trials did not report subgroup effects that enabled meaningful, adequately powered, comparisons.

Sensitivity analysis

We undertook sensitivity analysis for review outcomes with missing data, a judgement of high risk of bias in any risk of bias domain, denominator restrictions or 'no intervention' comparators. Where exclusion of studies left only a single study, we did not perform a separate sensitivity meta-analysis but reported the remaining study descriptively.

Certainty of the evidence assessment

Following standard Cochrane methodology, we created a summary of findings table to summarise our key results. We selected

our comparison of smoking counselling plus NRT with follow-up support versus usual care for the summary of findings table, and presented the outcomes of abstinence at six months and serious adverse events.

Two review authors (SP, AG) independently assessed our certainty in the body of evidence presented in the review using the five GRADE considerations (study limitations, consistency of effect, imprecision, indirectness, and publication bias) and overall RoB 2 judgements. We used methods and recommendations described in Section 12.2.1 of the *Cochrane Handbook* [65], employing GRADEpro GDT software [71]. We justified all decisions to downgrade or upgrade our level of certainty in the evidence using footnotes. We resolved disagreements about the judgements through discussion.

Consumer involvement

Consumer input informed the development of the review protocol. We have selected core outcomes for the review that ensure transparency, relevance, and comparability.

RESULTS

Description of studies

Results of the search

As described in [Electronic searches](#), we searched all databases from inception until 14 October 2024 and reran the search on 10 February 2026. Our search identified a total of 6328 records. After identification of ineligible studies using automation tools (Covidence) and removal of duplicates, 2097 records remained for title and abstract screening. We ruled out 2042 records at this stage, leaving 55 studies for full-text screening. From this we excluded 43 studies. A total of 10 studies and two ongoing studies were identified for inclusion in the review. The 'Characteristics of excluded studies' summarises the reasons for exclusion ([Supplementary material 3](#)). See [Figure 1](#) for the PRISMA diagram summarising the study selection process.

Included studies

An overview of included studies and which analyses they contribute to is provided in the 'Overview of included studies and synthesis table' [Table 1](#).

Study participants and settings

The 10 included studies consisted of nine RCTs and one cluster-randomised RCT representing 2262 participants. The included studies were conducted in the USA (three studies), Australia (two studies), Taiwan (two studies), Iran (two studies), and Israel (one study) in psychiatry inpatient units with adult participants.

Studies varied in inpatient settings: four studies recruited from acute inpatient psychiatric units (Brown 2021; Chan 2020 [72]; Metse 2017 [73, 74, 75, 76]; Prochaska 2014 [77, 78]); three studies from acute and longer stay units (Gelkopf 2012 [79]; Hickman 2015 [80, 81]; Tavakoli-Ardakani 2023 [82, 83]); one study from a longer stay unit (Chen 2013 [84]); and one study from both acute and dual diagnosis units (substance use and mental illness) (Stockings 2014 [85, 86]). One study did not describe unit type but was restricted to participants diagnosed with schizophrenia (Akbarpour 2010 [87]). Two included studies recruited participants diagnosed only with

schizophrenia-type disorders (Chen 2013; Gelkopf 2012). One study was based in a private psychiatric facility (Brown 2021).

Overall, there was a preponderance of males in the included studies with 1466 males compared to 776 females and 9 transgender participants. Across the included studies, Akbarpour 2010 had the highest proportion of male participants (100%), followed by Chen 2013 (92.9%), Tavakoli-Ardakani 2023 (87.2%), Chan 2020 (77.7%), Gelkopf 2012 (69.8%), Hickman 2015 (65.0%), Metse 2017 (61.3%), Prochaska 2014 (59.8%), Stockings 2014 (53.7%), and Brown 2021 with the lowest proportion (49.0%).

Most studies did not specify whether participants intended to quit smoking (Akbarpour 2010; Brown 2021; Chan 2020; Chen 2013; Metse 2017; Stockings 2014). Two trials explicitly enrolled participants regardless of quit intention, while two others required either willingness to engage in a smoking-reduction programme (Gelkopf 2012) or motivation to quit (Tavakoli-Ardakani 2023).

Interventions and comparators

Five studies compared interventions that offered motivational-type tobacco cessation counselling with pharmacological treatment (NRT) for withdrawal, and continued follow-up community support after discharge, in addition to usual-care treatments (Brown 2021; Hickman 2015; Metse 2017; Prochaska 2014; Stockings 2014). Usual care typically involved offering NRT for withdrawal plus brief advice to quit and/or self-help materials. One study noted usual care could also provide referral to Quitline/smoking support service and limited NRT on discharge, but this was limited and often variable (Stockings 2014). Two studies provided brief motivational interviews (10 to 15 minutes) plus tailored self-help literature (Metse 2017; Stockings 2014); two delivered a transtheoretical-model motivational tobacco cessation treatment (tailored to stage of change; 15 to 30 minutes) that included a computer-assisted session and treatment manual (Hickman 2015; Prochaska 2014); and one study delivered a 40-minute motivational interview counselling session tailored for people with severe mental illness (Brown 2021). All studies provided post-discharge support ranging between 8 and 12 weeks of either combination or monotherapy NRT and 12 to 26 weeks of psychosocial follow-up support, delivered via telecounselling (Metse 2017; Stockings 2014), automated messaging (Brown 2021), or computer-assisted contact (Hickman 2015; Prochaska 2014). The computer-assisted contacts for two studies repeated the in-hospital support, 'remembering' previous responses from participants (Hickman 2015; Prochaska 2014). Three studies also offered referral to Quitline (Brown 2021; Metse 2017; Stockings 2014). We classified these interventions as having level 4 intensity, defined as any hospital contact plus follow-up contacts lasting for more than a month, using Rigotti's hospital smoking cessation counselling intensity rating hierarchy [23].

Only one study evaluated a group behavioural intervention that was aimed at smoking reduction, comparing this to a waitlist control group who received a single lecture on the harms of smoking (Gelkopf 2012). Four studies compared pharmacological interventions, with two considering smoking medications, including bupropion versus placebo (Akbarpour 2010) and cytosine versus NRT gum (Tavakoli-Ardakani 2023). The two remaining studies examined NRT types and dose. One of these studies delivered six sessions of healthy lifestyle psychoeducation to all participants and compared NRT patch

dose levels (high versus low) (Chen 2013). The other study had a smoking reduction focus and provided all participants with weekly healthy lifestyle psychoeducation comparing different NRT types (patches, gum, and inhalers) to each other (Chan 2020). None of the group or pharmacological trials offered support after discharge. Consequently, we classified them as delivering an intensity level 2 intervention, defined as one or more in-hospital contacts that include counselling and last in total more than 15 minutes with no post-discharge contact [23].

Critical outcome measures

Five studies reported biochemically validated 7-day point-prevalence abstinence at six months (Brown 2021; Hickman 2015; Metse 2017; Prochaska 2014; Stockings 2014). Two studies also reported sustained abstinence, with one reporting one-month sustained abstinence (Metse 2017) and the other reporting continuous abstinence from date of discharge (Stockings 2014). One study provided non-verified self-reported abstinence at six months (Tavakoli-Ardakani 2023).

Important outcome measures

For secondary outcomes, five studies reported biochemically verified abstinence at end of intervention: Stockings 2014 (four months); Prochaska 2014 (three months); Hickman 2015 (three months); Chen 2013 (eight weeks); with one study reporting exhaled breath CO level at end of intervention (eight weeks) (Chan 2020). Post-discharge uptake of cessation supports (counselling or pharmacotherapy) was reported in three studies, two measuring intervention-only smoking cessation supports (Metse 2017; Stockings 2014), and one smoking cessation support from any source (Brown 2021). For the Metse 2017 study, this outcome was assessed at one-month in a companion publication from the same trial and used for this review (76). Six studies reported on change in cigarettes per day (CPD) at end of intervention: two at six months (Metse 2017; Stockings 2014); one at three months (Gelkopf 2012); and three at eight weeks (Akbarpour 2010; Chen 2013; Tavakoli-Ardakani 2023). Validated quality of life was reported in only one study (Gelkopf 2012).

Three studies reported adverse events for pharmacological interventions (Chan 2020; Chen 2013; Tavakoli-Ardakani 2023). Weight change was reported in two studies (Chan 2020; Gelkopf 2012). Four studies reported changes in psychiatric symptom severity (Chan 2020; Chen 2013; Gelkopf 2012; Stockings 2014). Five studies reported serious adverse events at the end of the trial (eight weeks to 18 months) (Brown 2021; Hickman 2015; Metse 2017; Prochaska 2014; Tavakoli-Ardakani 2023), and three studies reported an absence of any serious adverse events during the study (Chan 2020; Chen 2013; Gelkopf 2012). Two studies provided rehospitalisation outcomes with differing definitions and follow-up periods (Hickman 2015; Prochaska 2014). One reported this as overnight psychiatric rehospitalisation (Prochaska 2014), while the other included any acute hospitalisation or emergency department psychiatric care (Hickman 2015).

Excluded studies

We excluded 43 trials with reasons for exclusion listed in the 'Characteristics of excluded studies' table (Supplementary material 3) and PRISMA diagram (Figure 1). The most common reason for exclusion was the intervention not being initiated in the inpatient psychiatry unit (ineligible setting). Several studies did not meet

criteria for a randomised control trial or did not report critical or important outcomes specified in our protocol. One study (Das 2017 [88]) extracted and combined data from two included studies (Prochaska 2014; Hickman 2015) to examine quit outcomes for a subsample of psychiatric inpatients with co-occurring substance use disorders. Consequently, we excluded this paper to prevent duplication.

Ongoing studies

We found two ongoing RCTs, which tested the application of digitally delivered interventions initiated in the inpatient psychiatry setting combined with pharmacological interventions and follow-up support post-discharge (Brown 2022 [89], Bennett 2020 [90]). We contacted the authors of both ongoing studies. One study was awaiting publication (Bennett 2020), and the other was still recruiting participants (Brown 2022). Neither study had data available for inclusion in this review. More details provided in [Supplementary material 4](#).

Risk of bias in included studies

Our risk of bias assessments for each outcome, including all domain judgements and support for judgements, are in [Supplementary material 5](#) and also alongside all meta-analysis forest plots. To access further detailed risk of bias assessment data, please use this link: <https://doi.org/10.48610/7043ee2>.

Critical outcome

Smoking abstinence at 6 months

We conducted risk of bias assessments for the critical outcome of smoking abstinence at six months for the five studies contributing to this outcome. We judged two studies to have some concerns (mainly due to lack of information on allocation concealment, awareness of intervention assignment amongst those delivering the intervention, and insufficient detail to rule out deviations from intended interventions) (Brown 2021; Prochaska 2014). We judged the remaining three studies at high risk of bias due to missing outcome data that were plausibly related to the true value of the smoking abstinence status (Hickman 2015; Metse 2017; Stockings 2014). Taken together, we judged the overall risk of bias to be high for the review outcome of smoking abstinence at six months.

Important outcomes

We also assessed the following important outcomes for risk of bias.

Smoking abstinence at end of intervention

Three included studies contributed to this outcome, two of which we judged to have high risk of bias due to missing outcome data (Hickman 2015; Stockings 2014). We judged the remaining study to have some concerns of risk of bias due to concerns about the randomisation process and potential for deviations from intended interventions (Prochaska 2014). Overall, we judged the risk of bias for this outcome to be high.

Change in the number of cigarettes consumed per day (CPD)

Most included studies received high-risk judgements in multiple domains for this outcome measure. We rated all studies at high risk of bias for the outcome measurement domain due to the self-report nature of the outcome, combined with no information on question wording, recall period, or data-collection procedures. There were additional concerns about four studies rated at high risk of bias for the randomisation process due to inadequate descriptions of allocation concealment and randomisation methods (Akbarpour 2010; Chen 2013; Gelkopf 2012; Tavakoli-Ardakani 2023); four judged high risk for selective reporting of results (Akbarpour 2010; Chen 2013; Gelkopf 2012; Tavakoli-Ardakani 2023); and four rated high risk due to substantial and potentially outcome-related missing data (Chen 2013; Metse 2017; Stockings 2014; Tavakoli-Ardakani 2023). Overall, we judged the body of evidence for this outcome to be at high risk of bias.

Serious adverse events

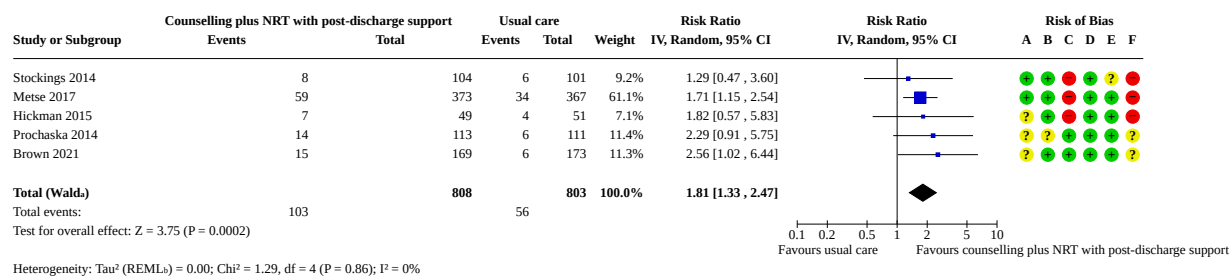
Four included studies contributed data for this outcome. We judged one to be at low risk of bias (Metse 2017) and the remaining three had some concerns (Brown 2021; Hickman 2015; Prochaska 2014), mainly due to a lack of information on allocation concealment, awareness of intervention assignment amongst those delivering the intervention, and insufficient detail to rule out deviations from intended interventions. Overall, we judged the risk of bias for serious adverse events as having some concerns.

Other potential sources of bias

We did not identify any other sources of bias. We did not assess potential publication bias or small-study effects as our analyses did not pool 10 or more studies [68].

Synthesis of results

See [Summary of findings 1](#) and risk of bias assessments ([Supplementary material 5](#)) for the critical outcome smoking abstinence at six months. All analyses are provided in [Supplementary material 6](#). We analysed pooled results with the random-effects model meta-analysis on studies sufficiently similar in statistical and clinical characteristics (see [Figure 2](#); [Figure 3](#); [Figure 4](#)). For studies or outcomes that could not be pooled, we conducted a narrative review using SWiM guidelines, as recommended in Chapter 12 of the *Cochrane Handbook for Systematic Reviews of Interventions* [91]. Outcome results are summarised narratively, with forest plots and summary tables provided where applicable (see [Table 2](#) for a summary of CPD outcomes across all included studies; [Figure 5](#) for the corresponding forest plot; [Table 3](#) for a summary of non-pharmacological adverse event outcomes). Planned subgroup analyses were not conducted due to insufficient data.

Figure 2. Analysis 1.1 Counselling plus NRT with post-discharge support versus Usual care 1: Smoking abstinence at 6 months**Footnotes**

aCI calculated by Wald-type method.

bTau² calculated by Restricted Maximum-Likelihood method.**Risk of bias legend**

(A) Bias arising from the randomization process

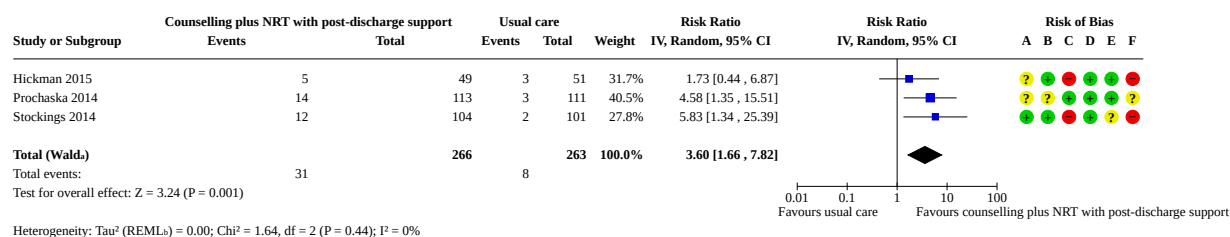
(B) Bias due to deviations from intended interventions

(C) Bias due to missing outcome data

(D) Bias in measurement of the outcome

(E) Bias in selection of the reported result

(F) Overall bias

Figure 3. Analysis 2.1 Counselling plus NRT with post-discharge support versus Usual care 1: Smoking abstinence at end of intervention**Footnotes**

aCI calculated by Wald-type method.

bTau² calculated by Restricted Maximum-Likelihood method.**Risk of bias legend**

(A) Bias arising from the randomization process

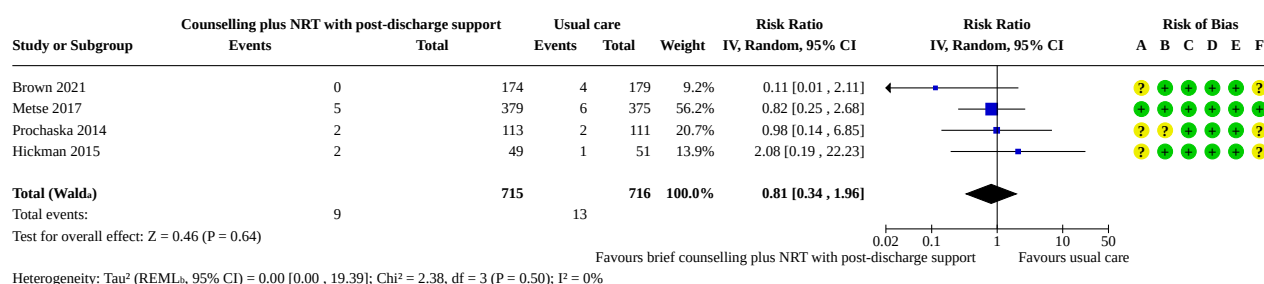
(B) Bias due to deviations from intended interventions

(C) Bias due to missing outcome data

(D) Bias in measurement of the outcome

(E) Bias in selection of the reported result

(F) Overall bias

Figure 4. Analysis 4.1 Counselling plus NRT with post-discharge support versus Usual care 1: Serious adverse events**Footnotes**

aCI calculated by Wald-type method.

bTau² calculated by Restricted Maximum-Likelihood method.**Risk of bias legend**

(A) Bias arising from the randomization process

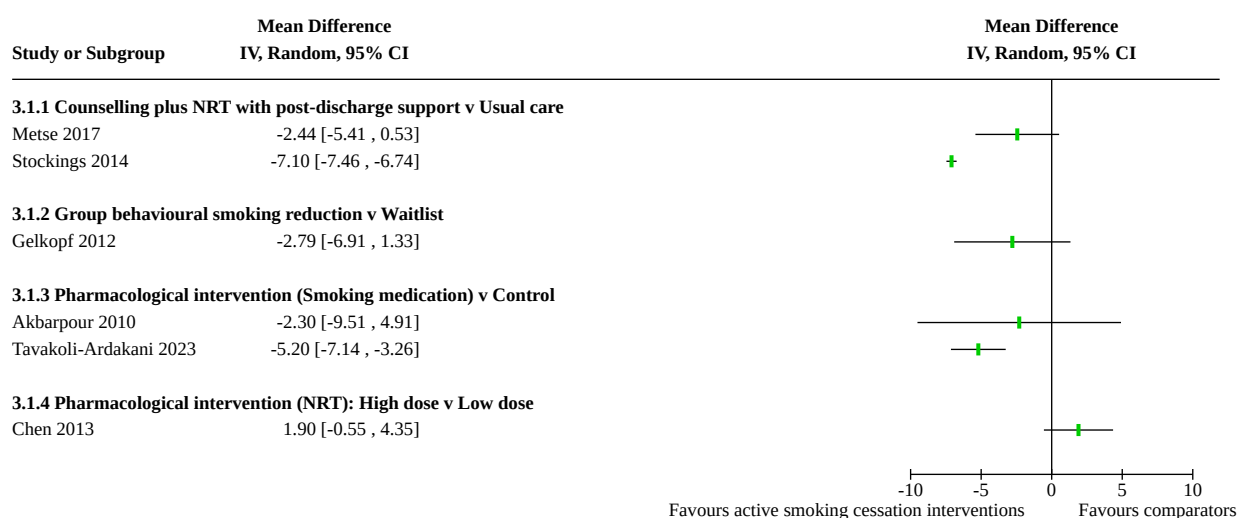
(B) Bias due to deviations from intended interventions

(C) Bias due to missing outcome data

(D) Bias in measurement of the outcome

(E) Bias in selection of the reported result

(F) Overall bias

Figure 5. Descriptive presentation of change in the number of cigarettes consumed per day (CPD)**1. Counselling plus pharmacotherapy (NRT) with post-discharge support intervention versus usual care****Critical outcome****Smoking abstinence at 6 months**

We pooled five studies (1611 participants) that offered counselling plus NRT and ongoing post-discharge cessation support, compared to usual care, and reported biochemically verified smoking point-prevalence abstinence at six months (Brown 2021; Hickman 2015; Metse 2017; Prochaska 2014; Stockings 2014). More participants successfully quit smoking at six months in the intervention group (RR 1.81, 95% CI 1.33 to 2.47; $I^2 = 0\%$; 5 studies, 1611 participants; low-certainty evidence; Figure 2; Analysis 1.1), with the estimate indicating a moderate benefit. However, high rates of missing data for this outcome in three of the five included studies limited our confidence in this effect, and we downgraded certainty of the

evidence to low. In a sensitivity analysis excluding the three studies at high risk of bias, the effect remained in favour of the intervention and was of greater magnitude (RR 2.42, 95% CI 1.26 to 4.65; P (probability statistic) < 0.001 , $I^2 = 0\%$; 2 studies, 566 participants). However, the small evidence base means these results should be interpreted with caution.

Two studies evaluating this comparison also reported sustained cessation rates. Stockings 2014 reported biochemically confirmed continuous abstinence at six months, with no significant differences reported between groups: intervention 0% versus control 1.9% ($P = 0.26$). Metse 2017 reported one-month prolonged abstinence at six months, finding no significant difference between groups: intervention 12.1% versus control 9.5% ($P = 0.37$).

Two studies compared the effect of psychiatric diagnosis on abstinence (Hickman 2015; Metse 2017). Metse 2017 reported no

differences in abstinence between psychotic compared to non-psychotic diagnoses at six months for 7-day point-prevalence abstinence (OR 1.01, 95% CI 0.91 to 1.13) or 1-month continued abstinence (odds ratio (OR) 1.04, 95% CI, 0.93 to 1.15). Hickman 2015 reported that the odds of abstinence over 12 months increased for people with diagnoses of psychotic disorders compared to unipolar depression (OR 5.16, 95% CI 1.76 to 15.07; $P = 0.03$).

Important outcomes

Smoking abstinence at end of intervention

A total of three studies (529 participants) contributed data for smoking abstinence at end of intervention for the comparison of counselling plus pharmacotherapy (NRT) with ongoing discharge support compared to usual care (Hickman 2015 and Prochaska 2014 at three months, and Stockings 2014 at four months). In this meta-analysis, more participants successfully quit smoking with the intervention before six months (RR 3.60, 95% CI 1.66 to 7.82; $I^2 = 0\%$; 3 studies, 529 participants; not specified; [Figure 3](#); Analysis 2.1), with intervention participants 3.6 times more likely to be abstinent than usual-care participants. However, we judged two of the three studies to have high risk of bias due to missing outcome data. Excluding these studies left only one study (224 participants), which showed an effect similar in direction to the primary analysis but very imprecise (RR 4.58, 95% CI 1.35 to 15.51).

Uptake of smoking cessation support post-intervention

Only three studies reported on post-discharge uptake of smoking cessation support; however, we did not pool data from the studies because of non-comparable definitions, timeframes, and data collection methods. Two studies measured uptake of post-discharge intervention supports delivered as part of the intervention at 1-month after discharge (Metse 2017; Stockings 2014). One study measured uptake of any smoking cessation treatment (e.g. counselling or pharmacotherapy) delivered from any source (Brown 2021). This study reported that uptake at six months was significantly more likely in the intervention group compared to the usual-care group (74.6% versus 40.5%; RR 1.8, 95% CI 1.51 to 2.25; $P < 0.001$) for both counselling (37.3% versus 11%) and pharmacotherapy (71% versus 37%) (Brown 2021). Of the two studies measuring post-discharge intervention delivered as part of the intervention, Stockings 2014 reported that approximately 90% of intervention participants (94/104) received at least some of the intervention cessation support, including 68% NRT (71/104), and Metse 2017 reported that 89% of participants accepted at least one counselling call and 79% used NRT. While the evidence consistently demonstrated high uptake of the intervention, variability in outcome definition and methodology, and no comparator, reduced our confidence in this finding.

Change in the number of cigarettes consumed per day (CPD)

Two studies comparing counselling plus pharmacotherapy (NRT) with ongoing discharge support versus usual care reported on cigarettes consumed per day (CPD) at end of intervention (Metse 2017; Stockings 2014). We did not pool results for this outcome due to the effect estimates of these studies differing substantially, resulting in high between-study heterogeneity that precluded meaningful meta-analysis. Therefore, we presented a forest plot for visual comparison rather than a pooled estimate ([Figure 5](#)). In these two studies, effect estimates favoured the intervention over the comparator, although the size of the change varied from a small

difference of approximately 2 CPD to a larger reduction of 7 CPD (see [Table 2](#)).

Quality of life measures (QoL)

None of the studies included in this comparison reported any quality of life measures.

Adverse events

Medication side effects

None of the studies in this comparison reported medication-related adverse events.

Psychiatric symptoms

Only one study in this group reported psychiatric symptoms, measuring psychological distress using the Kessler Psychological Distress Scale – 10 (K10) (Stockings 2014). The study found no significant differences in psychological distress between the intervention and usual care group at six months (see [Table 3](#)).

Serious adverse events (SAE)

Four studies (1431 participants) evaluating this comparison reported deaths. Two studies reported at over 12 months (Hickman 2015; Metse 2017), one at six months (Brown 2021), and one at 18 months (Prochaska 2014). We pooled the data and found that there were fewer deaths reported in the groups receiving counselling plus NRT with post-discharge cessation support groups than in the usual care groups, but this evidence is very uncertain (RR 0.81, 95% CI 0.34 to 1.96; $I^2 = 0\%$; 4 studies, 1431 participants; [Figure 4](#); Analysis 4.1; very low certainty evidence). Only 22 deaths occurred across the studies, and the confidence interval is wide and includes both the possibility of benefit and the possibility of harm.

Two studies in this comparison also reported rehospitalisation but used varied definitions. One measured psychiatry-only readmissions over 18 months, reporting greater rehospitalisation in the usual care group (OR 1.92, 95% CI 1.06 to 3.49; $P = 0.031$) (Prochaska 2014). The second study applied a broader definition of rehospitalisation that included any type of acute hospitalisation or emergency department psychiatric care, and reported that there was no difference between intervention and usual care at 12 months (55% usual care versus 57% intervention) (Hickman 2015).

2. Group behavioural smoking reduction programme versus waitlist

Gelkopf 2012 compared a group behavioural intervention to reduce the number of cigarettes smoked per day versus a waitlist control.

Critical outcome

Smoking abstinence at 6 months

Gelkopf 2012 did not report on our critical outcome of abstinence at six months.

Important outcomes

Change in the number of cigarettes consumed per day (CPD)

Gelkopf 2012 reported a statistically significant reduction in CPD at three months, with a repeated-measures ANOVA showing a significant differential change over time between groups. The intervention group reduced smoking by an average of 7.14 (SD 6.6) CPD compared to 2.00 (SD 4.5) in the waitlist group ($F(1,51) = 9.2$, P

< 0.05 ; $h^2 = 0.15$). See [Figure 5](#) and [Table 2](#). Participants smoking 11 or more CPD at baseline were approximately 10 times more likely to markedly reduce CPD at three months compared to those smoking 10 or less CPD at baseline, as shown by a significant χ^2 test ($\chi^2 = 23.36$, $P < 0.001$).

Quality of life measures (QoL)

Gelkopf 2012 reported no significant difference in quality of life scores between the intervention and waitlist groups at three months. Quality of life was measured using the Quality of Life Enjoyment and Satisfaction Questionnaire (Q-LES-Q) (0 to 100 scale) [92]. Scores changed from 62.23 to 60.71 in the intervention group and from 57.78 to 57.19 in the waitlist group, with a repeated-measures ANOVA showing no significant group difference ($F(1,51) = 0.13$, $P = 0.72$).

Adverse events

Adverse events were examined across two domains (psychiatric symptoms and weight change) and are summarised in [Table 3](#).

Psychiatric symptoms

Gelkopf 2012 used the Positive and Negative Syndrome Scale (PANSS), a clinician-rated measure of schizophrenia-related psychopathology [93]. Gelkopf 2012 found no significant increase in depression between the group behavioural smoking reduction intervention and waitlist control, and reported greater reductions in smoking associated with improvements in psychopathology symptoms, as indicated by a significant three-way interaction between time, group, and PANSS score ($F = 3.15$, $P = 0.05$). The intervention group also showed a significant reduction in PANSS scores at three months (63.87 ± 21.1 to 55.41 ± 21.1) compared to no change in the waitlist group (78.25 ± 26.5 to 79.58 ± 26.6 ; $P < 0.05$).

Weight change

No significant differences in weight change over the three-month follow-up period were observed between the intervention and waitlist groups.

Serious adverse events (SAE)

No serious adverse events were reported.

3. Pharmacotherapy: smoking medications versus placebo/NRT

Critical outcome

Smoking abstinence at six months

No studies evaluating smoking medications reported smoking abstinence at six months. Tavakoli-Ardakani 2023 provided self-reported, unverified abstinence at six months in a comparison of cytisine and NRT gum, finding low overall abstinence and no significant differences between conditions (cytisine 3/17, NRT 2/30; $P = 0.27$).

Important outcomes

Smoking abstinence at end of intervention

Neither study in this comparison group reported on this outcome.

Uptake of smoking cessation support post intervention

Neither study in this comparison group reported on this outcome.

Change in the number of cigarettes consumed per day (CPD)

Tavakoli-Ardakani 2023 and Akbarpour 2010 reported CPD at end of intervention (8 weeks) (see [Table 2](#)). We did not pool these results due to a high risk of bias across each study and clinical and methodological heterogeneity. A forest plot is provided for visual comparison rather than a pooled estimate ([Figure 5](#)).

Akbarpour 2010 reported a significant reduction in CPD at eight weeks favouring bupropion over placebo (11.1 ± 8.8 versus 13.4 ± 11.8), with bupropion reducing CPD by 2.3 more than placebo. Tavakoli-Ardakani 2023 reported CPD at 8 weeks with cytisine reducing 5.2 CPD more than NRT (5.2 ± 3.4 versus 10.4 ± 3.0), though this difference was not statistically significant ($P = 0.62$). Tavakoli-Ardakani 2023 also reported CPD reduction at six months with cytisine reducing CPD by 4.4 more than NRT (1.8 ± 1.6 versus 7.4 ± 3.4), and this difference was statistically significant ($P = 0.011$).

Quality of life measures (QoL)

None of the studies reported any quality of life measures.

Adverse events

Adverse events were examined across two domains (psychiatric symptoms and weight change) and are summarised in [Table 3](#).

Medication side effects

Akbarpour 2010 did not report on adverse events or side effects. Tavakoli-Ardakani 2023 reported more adverse events (e.g. nausea, vomiting, sleeping disorders) in the NRT group compared to cytisine, but no significant difference between conditions ($P = 0.14$).

Psychiatric symptoms

Akbarpour 2010, the bupropion study, assessed cognitive impairment using the Mini Mental State Examination abbreviated version (MMSE), finding significant improvement in MMSE scores for the bupropion group (bupropion: 23.5 (4.4) to 26.3 (3.5); $P = 0.002$) compared to no change in the placebo group (placebo: 22.8 (4.6) to 22.8 (3.5); $P = 0.954$) (see [Table 3](#)). However, the MMSE is a global and imprecise measure of cognitive impairment [94].

Serious adverse events (SAE)

Akbarpour 2010 did not report on serious adverse events. Tavakoli-Ardakani 2023 reported that there were no serious adverse events in either the cytisine or the NRT group.

4. Pharmacotherapy: nicotine replacement therapy (NRT) (different forms or doses)

Critical outcome

Smoking abstinence at six months

No studies in this comparison group reported on our critical outcome.

Important outcomes

Smoking abstinence at end of intervention

The two studies examining NRT interventions reported on smoking abstinence at end of intervention but were too methodologically

heterogeneous to combine in meta-analysis (Chan 2020; Chen 2013).

Chen 2013 measured 7-day point-prevalence biochemically verified abstinence at eight weeks, reporting a total of five participants abstinent (high dose 1/92, low dose 4/92) and no difference in abstinence between groups ($P = 0.174$).

Chan 2020 used CO₂ exhaled breath to measure smoking reduction across three NRT forms (inhaler, gum, patches), reporting that only the inhaler group demonstrated a significant reduction at eight weeks ($P = 0.032$) and significant decreasing trend in CO₂ levels over the eight-week period ($P = 0.002$).

Uptake of smoking cessation support post-intervention

Neither study reported smoking cessation support uptake after the intervention.

Change in the number of cigarettes consumed per day (CPD)

Only Chen 2013 measured CPD, reporting reductions in both groups over time ($P = 0.001$), with no significant difference in smoking reduction between the high- and low-dose NRT ($P = 0.35$). Although the low-dose group averaged 1.9 CPD fewer than the high-dose group at eight weeks (10.1 ± 7.7 versus 12.0 ± 9.2), we judged this to be a clinically unimportant difference (see [Figure 5](#) and [Table 2](#)).

Quality of life measures (QoL)

Neither of the studies in this comparison group measured quality of life.

Adverse events

Adverse events were examined across three domains (medication side effects, psychiatric symptoms, and weight change) and are summarised [Table 3](#).

Medication side effects

In the comparison of high-dose versus low-dose patches, Chen 2013 reported that five participants withdrew due to side effects (high dose 2/92, low dose 3/92), but Chen 2013 did not specify what the side effects were or their severity.

Chan 2020 reported no adverse reactions in any of the three NRT conditions.

Psychiatric symptoms

Chan 2020 used a global measure of psychiatric symptoms and reported significant improvement in psychiatric symptoms in all three NRT groups at eight weeks, with the inhaler group demonstrating significantly greater improvements than the patch group on post hoc comparisons ($F = 17.019$, $P < 0.001$). Chen 2013 reported small reductions in psychiatric symptoms in both high- and low-dose groups at eight weeks (PANSS change: -2.5 ± 11.0 in each group), with no significant difference between groups, indicating that both doses had a similar modest effect on psychiatric symptoms rather than no effect.

Weight change

Chan 2020 reported no significant body weight change in any of the three NRT groups ($P = 0.08$) at end of intervention (8 weeks).

Serious adverse events (SAE)

No serious adverse events were reported in either study.

DISCUSSION

Summary of main results

This review included 10 RCTs: five studies compared counselling plus nicotine replacement therapy and post-discharge support to usual care, one study compared a group smoking reduction programme to waitlist, and four studies investigated pharmacological interventions. The latter included smoking medications (bupropion versus placebo and cytosine versus NRT) and different NRT types/doses (inhaler versus gum versus patch and high-dose versus low-dose patches).

We found low-certainty evidence that offering smoking cessation counselling plus NRT during a psychiatry hospital admission and continuing this support post-discharge may increase smoking abstinence at six months compared with usual care. In a meta-analysis of five trials (1161 participants), 13 participants per 100 achieved abstinence at six months compared with 7 per 100 participants in usual care (absolute increase of 6 per 100). However, we rated the certainty of evidence as low because of the high risk of bias from missing outcome data that reduced our confidence in the effect estimate. Our sensitivity analysis suggested that the observed benefit was not entirely explained by the high-risk studies, since the effect persisted and increased when these studies were removed. Nonetheless, the limited number of remaining studies mean that the finding remains uncertain.

We found low-certainty evidence that interventions offering smoking cessation counselling plus NRT with post-discharge support may increase the number of people with smoking abstinence at the end of intervention when compared to usual care. In pooled analysis of three trials (529 participants), 11 participants per 100 achieved abstinence with the intervention compared to 3 per 100 participants with usual care. However, uncertainty remains around the precise magnitude of this effect due to a wide confidence interval and high risk of bias from missing outcome data in two of the three studies.

We found very low certainty evidence (based on four studies that involved 1431 participants) that smoking-cessation counselling plus NRT with post-discharge support may lead to a slight reduction in deaths compared with usual care (when measured 6 to 18 months after the start of the intervention or hospital discharge). However, the small number of events (22) and study limitations mean the true effect remains highly uncertain.

The small number of pharmacological interventions we identified for this review were too heterogeneous for quantitative pooling, highlighting the limited availability of randomised controlled trial evidence for pharmacological smoking cessation interventions initiated in the psychiatric setting. For smoking medications, cytosine showed some benefit over NRT gum, while bupropion showed little difference from placebo. However, both studies were methodologically weak, which limits our confidence in these findings. Across studies comparing different doses and types of NRT, smoking abstinence was low overall.

In terms of other important outcomes that we were unable to meta-analyse, we found that smoking cessation interventions

generally favoured a reduction in the number of cigarettes smoked per day when compared to control conditions. However, the high risk of bias in this outcome across studies and marked variability in reported reductions limited our confidence in these findings. While adverse event reporting was inconsistent across the included studies, medication-related events were generally limited to common side effects (e.g. nausea, vomiting) related to pharmacological interventions (NRT and cytisine), and there was no evidence across studies to suggest smoking cessation interventions worsened psychiatric symptoms or led to weight gain.

Limitations of the evidence included in the review

Few RCTs have been conducted that initiate smoking cessation interventions in the psychiatric inpatient setting, which constrains comparisons of different intervention types. Methodological and clinical heterogeneity across studies, including differences in interventions, outcome definitions and reporting time-points, alongside high risk of bias from missing data, often limited the comparability of outcomes. Moreover, larger sample sizes are needed.

There were few pharmacological trials and none using varenicline, which has shown promising outcomes in psychiatric populations in community-based trials [27, 95]. Where pharmacological studies were available, there were methodological limitations and follow-up abstinence measures were rare, making longer-term abstinence for inpatient-initiated therapy uncertain. As no eligible e-cigarette trials met the inclusion criteria, a formal review of this intervention was not possible.

Across all included trials, there was a higher proportion of male participants; however, the studies contributing to our meta-analyses were balanced in terms of sex distribution. Nonetheless, this imbalance warrants caution when generalising findings to female inpatient psychiatry populations. Across studies with data that we were able to pool, intervention components varied, including NRT delivery, which ranged from combination regimes to monotherapy. Few studies reported psychiatric symptom severity, and reporting of adverse events and rehospitalisation was inconsistent. Only two studies examined effects by diagnosis, and none reported these in enough detail to compare cessation outcomes across diagnostic groups.

Limitations of the review processes

We consider the review process we used to be robust. We followed best practice for conducting Cochrane reviews, including duplicate processes and a search strategy that included expansive search terms, and a thorough search of published and unpublished sources developed by a search specialist (GV). It is, however, possible that there are unpublished data that our searches did not find due to indexing limitations or use of alternative terminology to that used in our search.

We used intention-to-treat analyses, as per the standard approach for missing data in this field, for our outcome analyses. This is the most conservative approach but may understate the true abstinence rate.

Agreements and disagreements with other studies or reviews

This is the first Cochrane review to systematically evaluate the effectiveness of smoking cessation interventions that actively support abstinence, as distinct from smoking bans or other harm-reduction approaches initiated in inpatient psychiatry. Accordingly, direct comparisons with prior reviews are not possible. However, below we contextualise our findings against relevant reviews from other settings or psychiatric populations.

Interventions that combine pharmacotherapy and behavioural support have been shown to improve cessation compared to usual or minimal care in a variety of settings and populations [96]. In general, for hospitalised patients, hospital-initiated interventions that combine counselling and NRT in hospital and are continued at least one-month post-discharge have been shown to increase quit rates compared to no intervention [22]. Our review extends the evidence base by providing low-certainty evidence that, in inpatient psychiatry, counselling plus NRT with ongoing cessation support post-discharge can improve abstinence rates compared to usual care. The consistency of effects across different settings and population groups strengthens confidence in the findings and their generalisability.

We identified only one RCT examining a group behavioural intervention compared to waitlist, which showed greater reduction in cigarettes smoked for the smoking reduction intervention. This is consistent with a systematic review of group behaviour therapy programmes for people who smoke that found group programmes were superior to self-help or other less intensive interventions, but they were not reported as consistently better than healthcare provider advice [97].

Reviews in psychiatry populations have demonstrated the effectiveness of smoking medications including bupropion and varenicline in community trials for people with schizophrenia and past depression [46, 47]. Hospital-initiated varenicline has also been reported to have moderate-certainty evidence for improving cessation outcomes, when compared to placebo or non-varenicline interventions in general hospital settings [22]. In our review, only two pharmacotherapy smoking medication trials were identified (bupropion, cytisine). Overall quit rates were low, and the studies had notable methodological limitations. We did not identify any varenicline RCTs initiated in the psychiatric inpatient setting. The absence of evidence does not indicate the ineffectiveness of pharmacological cessation treatments; rather, it reflects the limited number of well-designed randomised controlled trials in this area and highlights the need for more inpatient RCTs to establish their effects in this setting. Particularly given the robust community-trial-based evidence supporting their safety and effectiveness in this population [26, 27, 38, 45, 48]. Previous research has also shown smoking cessation does not worsen mental health and may in fact improve depression, anxiety, stress, and mood in the longer term [39, 44, 98]. Consistent with this broader evidence base, our review found that amongst studies assessing psychiatric symptoms, none reported any deterioration.

AUTHORS' CONCLUSIONS

Implications for practice

There is low-certainty evidence that providing counselling plus nicotine replacement therapy on psychiatry inpatient wards and continuing this support post-discharge may improve abstinence rates six months later compared to usual care. Across the evidence reviewed, we observed no worsening of psychiatric symptoms or increased risk of adverse events. There was also very low-certainty evidence that these interventions may reduce mortality, although the evidence was limited and highly uncertain.

From a public health perspective, the available evidence indicates potential benefit in offering smoking cessation intervention during psychiatry inpatient admission and maintaining this support after discharge. Implementing this effectively would require embedding smoking cessation care into routine service delivery and addressing practical considerations such as staff training, medication availability, discharge planning, and the delivery of post-discharge supports.

Implications for research

Psychiatry inpatient populations are often under-represented in smoking cessation research. This review shows they can be recruited and retained, and can achieve cessation, but the current evidence base remains limited and heterogeneous. Priorities for future research include high-quality randomised controlled trials that evaluate pharmacological interventions (e.g. varenicline, cytisine) initiated in the inpatient setting and that assess longer-term abstinence outcomes. To strengthen comparability across studies and interpretability of pooled results, standardised outcomes including biochemically verified abstinence, validated psychiatric symptom scales, and consistent reporting of adverse events are needed. Exploring differences by diagnosis, sex/gender, delivery modality, and follow-up duration would improve the available evidence. Two ongoing trials identified in this review include digitally delivered interventions that may further clarify some of these effects to broaden this evidence base. Notably, no e-cigarette trials were identified for inclusion in this review, representing a research gap. Finally, embedding co-design with consumers, carers, and clinicians could enhance participation, increase engagement, and improve translation of research findings to real-world practice.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD015934.pub2](https://doi.org/10.1002/14651858.CD015934.pub2).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Risk of bias

Supplementary material 6 Analyses

Supplementary material 7 Data package

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

The following people conducted the editorial process for this article.

- Sign-off Editor (final editorial decision): Jamie Hartmann-Boyce, University of Massachusetts Amherst
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Hannah Payne, Cochrane Central Editorial Service
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Contributions of authors

Sally Plevier led the conceptualisation, methodology, writing of original draft, writing of manuscript and editing, study selection, data extraction, risk of bias assessment, GRADE assessment, and overall project administration (lead).

Steve Kisely contributed to the conceptualisation, methodology, validation, writing of manuscript and editing, and supervision.

Billie Bonevski contributed to the conceptualisation, methodology, writing of manuscript and editing, and supervision.

Dan Siskind contributed to the conceptualisation, methodology, writing of manuscript and editing, and supervision.

Ashleigh Guillaumier contributed to the conceptualisation and editing of the manuscript, and provided study selection, data extraction, risk of bias assessment, and GRADE assessment.

Parth Thaker contributed to the conceptualisation and editing of the manuscript, and provided study selection and risk of bias assessment.

Julia Yamazki-Tan contributed to the conceptualisation and reviewing of the manuscript, and provided study selection and risk of bias assessment.

Greta Vos developed the database search strategy, investigation, and contributed to the editing of the manuscript.

Doyoon Kim contributed to reviewing the manuscript and provided risk of bias assessment.

Coral Gartner contributed to the conceptualisation, methodology, validation, study selection, risk of bias assessment, GRADE assessment, manuscript editing, and supervision.

Declarations of interest

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Author SK declares working as a healthcare professional with no present or recent affiliations, non-financial, financial or other involvement in any organisation or entity with an interest in the review's findings that might lead to a real or perceived conflict of interest.

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Data, code and other materials

As part of the published Cochrane review, the following has been made available for download for users of the Cochrane Library: full search strategies for each database ([Supplementary material 1](#)); full citations of each unique report for all studies (included, ongoing or awaiting classification, or excluded at the full-text screen) in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows ([Supplementary material 2](#); [Supplementary material 3](#); [Supplementary material 4](#); [Supplementary material 5](#)). Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods ([Supplementary material 6](#); [Supplementary material 7](#)). Template data extraction forms from Excel are available from the authors on reasonable request (<https://doi.org/10.48610/7043ee2>).

REFERENCES

1. Brown S, Kim M, Mitchell C, Inskip H. Twenty-five year mortality of a community cohort with schizophrenia. *British Journal of Psychiatry* 2010;**196**(2):116-21. [DOI: [10.1192/bjp.bp.109.067512](https://doi.org/10.1192/bjp.bp.109.067512)]
2. Dickerson F, Stallings C, Origoni A, Schroeder J, Khushalani S, Yolken R. Mortality in schizophrenia: clinical and serological predictors. *Schizophrenia Bulletin* 2014;**40**(4):796-803. [DOI: [10.1093/schbul/sbt113](https://doi.org/10.1093/schbul/sbt113)]
3. Erlangsen A, Andersen PK, Toender A, Laursen TM, Nordentoft M, Canudas-Romo V. Cause-specific life-years lost in people with mental disorders: a nationwide, register-based cohort study. *Lancet Psychiatry* 2017;**4**(12):937-45.
4. Afzal M, Siddiqi N, Ahmad B, Afsheen N, Aslam F, Ali A, et al. Prevalence of overweight and obesity in people with severe mental illness: systematic review and meta-analysis. *Frontiers in Endocrinology (Lausanne)* 2021;**12**:769309.
5. Lawn S, Campion J. Achieving smoke-free mental health services: lessons from the past decade of implementation research. *International Journal of Environmental Research and Public Health* 2013;**10**(9):4224-44. [DOI: [10.3390/ijerph10094224](https://doi.org/10.3390/ijerph10094224)]
6. Rogers ES. Financial distress and smoking-induced deprivation in smokers with depression. *American Journal of Health Behavior* 2019;**43**(1):219-27. [DOI: [10.5993/AJHB.43.1.18](https://doi.org/10.5993/AJHB.43.1.18)]
7. Dickerson F, Schroeder J, Katsafanas E, Khushalani S, Origoni AE, Savage C, et al. Cigarette smoking by patients with serious mental illness, 1999-2016: an increasing disparity. *Psychiatric Services* 2018;**69**(2):147-53. [DOI: [10.1176/appi.ps.201700118](https://doi.org/10.1176/appi.ps.201700118)]
8. Fornaro M, Carvalho AF, De Prisco M, Mondin AM, Billeci M, Selby P, et al. The prevalence, odds, predictors, and management of tobacco use disorder or nicotine dependence among people with severe mental illness: systematic review and meta-analysis. *Neuroscience & Biobehavioral Reviews* 2021;**132**:289-303. [DOI: [10.1016/j.neubiorev.2021.11.039](https://doi.org/10.1016/j.neubiorev.2021.11.039)]
9. Lawrence D, Mitrou F, Zubrick SR. Smoking and mental illness: results from population surveys in Australia and the United States. *BioMed Central Public Health* 2009;**9**:285. [DOI: [10.1186/1471-2458-9-285](https://doi.org/10.1186/1471-2458-9-285)]
10. Cooper J, Mancuso SG, Borland R, Slade T, Galletly C, Castle D. Tobacco smoking among people living with a psychotic illness: the second Australian Survey of Psychosis. *Australian and New Zealand Journal of Psychiatry* 2012;**46**(9):851-63. [DOI: [10.1177/0004867412449876](https://doi.org/10.1177/0004867412449876)]
11. Richardson S, McNeill A, Brose LS. Smoking and quitting behaviours by mental health conditions in Great Britain (1993-2014). *Addictive Behaviors* 2019;**90**:14-19. [DOI: [10.1016/j.addbeh.2018.10.011](https://doi.org/10.1016/j.addbeh.2018.10.011)]
12. Smith PH, Chhipa M, Bystrick J, Roy J, Goodwin RD, McKee SA. Cigarette smoking among those with mental disorders in the US population: 2012-2013 update. *Tobacco Control* 2020;**29**(1):29-35. [DOI: [10.1136/tobaccocontrol-2018-054268](https://doi.org/10.1136/tobaccocontrol-2018-054268)]
13. Lineberry TW, Allen JD, Nash J, Galardy CW. Population-based prevalence of smoking in psychiatric inpatients: a focus on acute suicide risk and major diagnostic groups. *Comprehensive Psychiatry* 2009;**50**(6):526-32. [DOI: [10.1016/j.comppsy.2009.01.004](https://doi.org/10.1016/j.comppsy.2009.01.004)]
14. Metse AP, Wiggers J, Wye P, Wolfenden L, Freund M, Clancy R, et al. Efficacy of a universal smoking cessation intervention initiated in inpatient psychiatry and continued post-discharge: a randomised controlled trial. *Australian and New Zealand Journal of Psychiatry* 2017;**51**(4):366-81. [DOI: [10.1177/0004867417692424](https://doi.org/10.1177/0004867417692424)]
15. Plevier S, McCarthy I, Anzolin M, Emmerson B, Allan J, Hay K. Queensland smoking care in adult acute mental health inpatient units: supporting practice change. *Australian and New Zealand Journal of Psychiatry* 2020;**54**(9):919-27. [DOI: [10.1177/0004867420917443](https://doi.org/10.1177/0004867420917443)]
16. De Leon J, Diaz FJ. A meta-analysis of worldwide studies demonstrates an association between schizophrenia and tobacco smoking behaviors. *Schizophrenia Research* 2005;**76**(2):135-57. [DOI: [10.1016/j.schres.2005.02.010](https://doi.org/10.1016/j.schres.2005.02.010)]
17. Mann-Wrobel MC, Bennett ME, Weiner EE, Buchanan RW, Ball MP. Smoking history and motivation to quit in smokers with schizophrenia in a smoking cessation program. *Schizophrenia Research* 2011;**126**(1-3):277-83. [DOI: [10.1016/j.schres.2010.10.030](https://doi.org/10.1016/j.schres.2010.10.030)]
18. Smith PH, Mazure CM, McKee SA. Smoking and mental illness in the U.S. population. *Tobacco Control* 2014;**23**(e2):e147-53. [DOI: [10.1136/tobaccocontrol-2013-051466](https://doi.org/10.1136/tobaccocontrol-2013-051466)]
19. Callaghan RC, Veldhuizen S, Jeysingh T, Orlan C, Graham C, Kakouris G, et al. Patterns of tobacco-related mortality among individuals diagnosed with schizophrenia, bipolar disorder, or depression. *Journal of Psychiatric Research* 2014;**48**(1):102-10. [DOI: [10.1016/j.jpsychires.2013.09.014](https://doi.org/10.1016/j.jpsychires.2013.09.014)]
20. Bonevski B, Borland R, Paul CL, Richmond RL, Farrell M, Baker A, et al. No smoker left behind: it's time to tackle tobacco in Australian priority populations. *Medical Journal of Australia* 2017;**207**(4):141-2. [DOI: [10.5694/mja16.01425](https://doi.org/10.5694/mja16.01425)]
21. Williams JM, Steinberg ML, Griffiths KG, Cooperman N. Smokers with behavioral health comorbidity should be designated a tobacco use disparity group. *American Journal of Public Health* 2013;**103**(9):1549-55. [DOI: [10.2105/AJPH.2013.301232](https://doi.org/10.2105/AJPH.2013.301232)]
22. Streck JM, Rigotti NA, Livingstone-Banks J, Tindle HA, Clair C, Munafò MR, et al. Interventions for smoking cessation in hospitalised patients. *Cochrane Database of Systematic Reviews* 2024, Issue 5. Art. No: CD001837. [DOI: [10.1002/14651858.CD001837.pub4](https://doi.org/10.1002/14651858.CD001837.pub4)]

23. Rigotti NA, Clair C, Munafo MR, Stead LF. Interventions for smoking cessation in hospitalised patients. *Cochrane Database of Systematic Reviews* 2012, Issue 5. Art. No: CD001837. [DOI: [10.1002/14651858.CD001837.pub3](https://doi.org/10.1002/14651858.CD001837.pub3)]
24. Fiore MC, Jaen CR, Baker TB, Bailey WC, Bennett G, Benowitz NL, et al. A clinical practice guideline for treating tobacco use and dependence: 2008 update. A U.S. Public Health Service report. *American Journal of Preventive Medicine* 2008;**35**(2):158-76. [DOI: [10.1016/j.amepre.2008.04.009](https://doi.org/10.1016/j.amepre.2008.04.009)]
25. The Royal Australian College of General Practitioners. Supporting Smoking Cessation: A Guide for Health Professionals. 2nd edition. East Melbourne, Vic: The Royal Australian College of General Practitioners, 2019. [URL: <https://www.racgp.org.au/clinical-resources/clinicalguidelines/key-racgp-guidelines/view-all-racgp-guidelines/supporting-smoking-cessation>]
26. Roberts E, Eden Evins A, McNeill A, Robson D. Efficacy and tolerability of pharmacotherapy for smoking cessation in adults with serious mental illness: a systematic review and network meta-analysis. *Addiction* 2016;**111**(4):599-612. [DOI: [10.1111/add.13236](https://doi.org/10.1111/add.13236)]
27. Siskind DJ, Wu BT, Wong TT, Firth J, Kisely S. Pharmacological interventions for smoking cessation among people with schizophrenia spectrum disorders: a systematic review, meta-analysis, and network meta-analysis. *Lancet Psychiatry* 2020;**7**(9):762-74. [DOI: [10.1016/S2215-0366\(20\)30261-3](https://doi.org/10.1016/S2215-0366(20)30261-3)]
28. Giulietti F, Filippini A, Rosettani G, Giordano P, Iacoacci C, Spannella F, et al. Pharmacological approach to smoking cessation: an updated review for daily clinical practice. *High Blood Pressure & Cardiovascular Prevention* 2020;**27**(5):349-62. [DOI: [10.1007/s40292-020-00396-9](https://doi.org/10.1007/s40292-020-00396-9)]
29. Caponnetto P, Polosa R. Approved and emerging smoking cessation treatments for people with schizophrenia spectrum disorders: a narrative review. *Health Psychology Research* 2020;**8**(2):9237. [DOI: [10.4081/hpr.2020.9237](https://doi.org/10.4081/hpr.2020.9237)]
30. Lindson N, Butler AR, McRobbie H, Bullen C, Hajek P, Wu AD, et al. Electronic cigarettes for smoking cessation. *Cochrane Database of Systematic Reviews* 2025, Issue 1. Art. No: CD010216. [DOI: [10.1002/14651858.CD010216.pub9](https://doi.org/10.1002/14651858.CD010216.pub9)]
31. Peckham E, Mishu M, Fairhurst C, Robson D, Bradshaw T, Arundel C, et al. E-cigarette use and associated factors among smokers with severe mental illness. *Addictive Behaviors* 2020;**108**:106456. [DOI: [10.1016/j.addbeh.2020.106456](https://doi.org/10.1016/j.addbeh.2020.106456)]
32. The Royal Australian and New Zealand College of Psychiatrists. Mental Health Clinician Guidance for Managing People's Smoking Cessation. Melbourne, Vic: The Royal Australian and New Zealand College of Psychiatrists, 2022.
33. Prochaska JJ, Hall SE, Delucchi K, Hall SM. Efficacy of initiating tobacco dependence treatment in inpatient psychiatry: a randomized controlled trial. *American Journal of Public Health* 2014;**104**(8):1557-65. [DOI: [10.2105/AJPH.2013.301403](https://doi.org/10.2105/AJPH.2013.301403)]
34. Siru R, Hulse GK, Tait RJ. Assessing motivation to quit smoking in people with mental illness: a review. *Addiction* 2009;**104**(5):719-33. [DOI: [10.1111/j.1360-0443.2009.02545.x](https://doi.org/10.1111/j.1360-0443.2009.02545.x)]
35. Brose LS, Simonavicius E, McNeill A. Maintaining abstinence from smoking after a period of enforced abstinence - systematic review, meta-analysis and analysis of behaviour change techniques with a focus on mental health. *Psychological Medicine* 2018;**48**(4):669-78. [DOI: [10.1017/S0033291717002021](https://doi.org/10.1017/S0033291717002021)]
36. Prochaska JJ, Fletcher L, Hall SE, Hall SM. Return to smoking following a smoke-free psychiatric hospitalization. *American Journal on Addictions* 2006;**15**(1):15-22. [DOI: [10.1080/10550490500419011](https://doi.org/10.1080/10550490500419011)]
37. Lawn S. Australians with mental illness who smoke. *British Journal of Psychiatry* 2001;**178**(1):85. [DOI: [10.1192/bjp.178.1.85](https://doi.org/10.1192/bjp.178.1.85)]
38. Beard E, Jackson SE, Anthenelli RM, Benowitz NL, St Aubin L, McRae T, et al. Estimation of risk of neuropsychiatric adverse events from varenicline, bupropion and nicotine patch versus placebo: secondary analysis of results from the EAGLES trial using Bayes factors. *Addiction* 2021;**116**(10):2816-2824. [DOI: [10.1111/add.15440](https://doi.org/10.1111/add.15440)]
39. Taylor GM, Lindson N, Farley A, Leinberger-Jabari A, Sawyer K, Te Water Naude R, et al. Smoking cessation for improving mental health. *Cochrane Database of Systematic Reviews* 2021, Issue 3. Art. No: CD013522. [DOI: [10.1002/14651858.CD013522.pub2](https://doi.org/10.1002/14651858.CD013522.pub2)]
40. Wu AD, Gao M, Aveyard P, Taylor G. Smoking cessation and changes in anxiety and depression in adults with and without psychiatric disorders. *JAMA Network Open* 2023;**6**(5):e2316111. [DOI: [10.1001/jamanetworkopen.2023.16111](https://doi.org/10.1001/jamanetworkopen.2023.16111)]
41. Huddleston L, Shoesmith E, Pervin J, Lorencatto F, Watson J, Ratschen E. A systematic review of mental health professionals, patients, and carers' perceived barriers and enablers to supporting smoking cessation in mental health settings. *Nicotine Tobacco Research* 2022;**24**(7):945-54. [DOI: [10.1093/ntr/ntac004](https://doi.org/10.1093/ntr/ntac004)]
42. Chen HK, Lan TH, Wu BJ. A double-blind randomized clinical trial of different doses of transdermal nicotine patch for smoking reduction and cessation in long-term hospitalized schizophrenic patients. *European Archives of Psychiatry and Clinical Neuroscience* 2013;**263**(1):75-82. [DOI: [10.1007/s00406-012-0338-3](https://doi.org/10.1007/s00406-012-0338-3)]
43. Khanna P, Clifton AV, Banks D, Tosh GE. Smoking cessation advice for people with serious mental illness. *Cochrane Database of Systematic Reviews* 2016, Issue 1. Art. No: CD009704. [DOI: [10.1002/14651858.CD009704.pub2](https://doi.org/10.1002/14651858.CD009704.pub2)]
44. Pinho S, Rocha V, Vieira-Coelho MA. Effectiveness of multimodal interventions focused on smoking cessation in patients with schizophrenia: a systematic review. *Schizophrenia Research* 2021;**231**:145-53. [DOI: [10.1016/j.schres.2021.03.012](https://doi.org/10.1016/j.schres.2021.03.012)]
45. Smith RC, Amiaz R, Si T-M, Maayan L, Jin H, Boules S, et al. Varenicline effects on smoking, cognition, and psychiatric symptoms in schizophrenia: a double-blind randomized

trial. *PLOS ONE* 2016;**11**(1):e0143490. [DOI: [10.1371/journal.pone.0143490](https://doi.org/10.1371/journal.pone.0143490)]

46. Tsoi DT, Porwal M, Webster AC. Interventions for smoking cessation and reduction in individuals with schizophrenia. *Cochrane Database of Systematic Reviews* 2013, Issue 2. Art. No: CD007253. [DOI: [10.1002/14651858.CD007253.pub3](https://doi.org/10.1002/14651858.CD007253.pub3)]

47. Van der Meer RM, Willemsen MC, Smit F, Cuijpers P. Smoking cessation interventions for smokers with current or past depression. *Cochrane Database of Systematic Reviews* 2013, Issue 8. Art. No: CD006102. [DOI: [10.1002/14651858.CD006102.pub2](https://doi.org/10.1002/14651858.CD006102.pub2)]

48. Peckham E, Brabyn S, Cook L, Tew G, Gilbody S. Smoking cessation in severe mental ill health: What works? An updated systematic review and meta-analysis. *BioMed Central Psychiatry* 2017;**17**(1):252. [DOI: [10.1186/s12888-017-1419-7](https://doi.org/10.1186/s12888-017-1419-7)]

49. Kagabo R, Gordon AJ, Okuyemi K. Smoking cessation in inpatient psychiatry treatment facilities: a review. *Addictive Behaviors Reports* 2020;**11**:100255. [DOI: [10.1016/j.abrep.2020.100255](https://doi.org/10.1016/j.abrep.2020.100255)]

50. Higgins JP, Lasserson T, Thomas J, Flemyng E, Churchill R. Methodological Expectations of Cochrane Intervention Reviews. Cochrane: London, August 2023. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/mecir-manual>.

51. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;**372**:71. [DOI: [10.1136/bmj.n71](https://doi.org/10.1136/bmj.n71)]

52. Stockings E, Black N, Bartlem KM, Metse AP, Regan T, Bailey JM, et al. Outpatient interventions for smoking cessation and reduction for adults with a mental disorder. *Cochrane Database of Systematic Reviews* 2019, Issue 3. Art. No: CD013286. [DOI: [10.1002/14651858.CD013286](https://doi.org/10.1002/14651858.CD013286)]

53. Evins AE, Cather C, Laffer A. Treatment of tobacco use disorders in smokers with serious mental illness: toward clinical best practices. *Harvard Review of Psychiatry* 2015;**23**(2):90-98.

54. Van Krugten FCW, Feskens K, Busschbach JJV, Hakkaart-van Roijen L, Brouwer WBF. Instruments to assess quality of life in people with mental health problems: a systematic review and dimension analysis of generic, domain- and disease-specific instruments. *Health and Quality of Life Outcomes* 2021;**19**(1):1-249.

55. Plevier S, Kisely SR, Bonevski B, Siskind D, Guillaumier A, McCarter K, et al. Interventions for smoking cessation in inpatient psychiatry settings. *Cochrane Database of Systematic Reviews* 2024, Issue 9. Art. No: CD015934. [DOI: [10.1002/14651858.CD015934](https://doi.org/10.1002/14651858.CD015934)]

56. Brown RA, Minami H, Hecht J, Kahler CW, Price LH, Kjome KL, et al. Sustained care smoking cessation intervention for individuals hospitalized for psychiatric disorders: the Helping HAND 3 randomized clinical trial. *JAMA Psychiatry* 2021;**78**(8):839-47. [DOI: [10.1001/jamapsychiatry.2021.0707](https://doi.org/10.1001/jamapsychiatry.2021.0707)]

57. Hecht J, Rigotti NA, Minami H, Kjome KL, Bloom EL, Kahler CW, et al. Adaptation of a sustained care cessation intervention for smokers hospitalized for psychiatric disorders: study protocol for a randomized controlled trial. *Contemporary Clinical Trials* 2019;**83**:18-26. [DOI: [10.1016/j.cct.2019.06.001](https://doi.org/10.1016/j.cct.2019.06.001)]

58. Review Manager (RevMan). Version 7.12.0. The Cochrane Collaboration, 2024. Available at <https://revman.cochrane.org>.

59. Covidence. Melbourne, Australia: Veritas Health Innovation, 2024. Available at <https://www.covidence.org>.

60. Microsoft Excel. Microsoft, accessed prior to 12 August 2024.

61. Heatherton TF, Kozolowski LT, Frecker RC, Fagerstrom, K. The Fagerstrom Test for Nicotine Dependence: a revision of the Fagerstrom Tolerance Questionnaire. *British Journal of Addiction* 1991;**86**(9):1119-27. [DOI: [10.1111/j.1360-0443.1991.tb01879.x](https://doi.org/10.1111/j.1360-0443.1991.tb01879.x)]

62. Heatherton TF, Kozlowski LT, Frecker RC, Rickert W, Robinson J. Measuring the heaviness of smoking: using self-reported time to the first cigarette of the day and number of cigarettes smoked per day. *British Journal of Addiction* 1989;**84**(7):791-9. [DOI: [10.1111/j.1360-0443.1989.tb03059.x](https://doi.org/10.1111/j.1360-0443.1989.tb03059.x)]

63. Higgins JP, Savović J, Page MJ, Elbers RG, Sterne JA. Chapter 8: Assessing risk of bias in a randomized trial. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.3 (updated February 2022). Cochrane, 2022. Available from <https://www.cochrane.org/handbook/archive/v6.3>.

64. Higgins JP, Li T, Deeks JJ editor(s). Chapter 6: Choosing effect measures and computing estimates of effect. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.3 (updated February 2022). Cochrane, 2022. Available from <https://www.cochrane.org/handbook/archive/v6.3>.

65. Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al editor(s). *Cochrane Handbook for Systematic Reviews of Interventions*. 2nd Edition. Chichester (UK): John Wiley & Sons, 2019. Available from <https://www.cochrane.org/handbook>.

66. Sterne JAC, Savović J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;**348**:366.

67. Higgins JPT, Savović J, Page MJ, Elbers RG, Sterne JAC. Chapter 8: Assessing risk of bias in a randomized trial. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.4 (updated August 2023). Cochrane, 2023. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v6.4>.

68. Higgins JP, Altman DG, Gotzsche PC, Juni P, Moher D, Oxman AD, et al; Cochrane Bias Methods Group, Cochrane Statistical Methods Group. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *British Medical Journal* 2011;**343**:d5928. [DOI: [10.1136/bmj.d5928](https://doi.org/10.1136/bmj.d5928)]

69. Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ* 2003;**327**(7414):557-60. [DOI: [10.1136/bmj.327.7414.557](https://doi.org/10.1136/bmj.327.7414.557)]
70. Campbell M, McKenzie JE, Sowden A, Katikireddi SV, Brennan SE, Ellis S, et al. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline. *BMJ* 2020;**368**:l6890.
71. GRADEpro GDT. Version accessed 27 April 2022. Hamilton (ON): McMaster University (developed by Evidence Prime), 2022. Available at <https://gradepr.org>.
72. Chan H-Y, Chen J-J, Pan Y-J, Hsu C-C. Nicotine replacement therapy and healthy lifestyle psychoeducation for smoking reduction in acute psychiatric inpatients: a cluster-randomized parallel study. *Journal of Clinical Psychopharmacology* 2020;**40**(2):149-56. [DOI: [10.1097/JCP.0000000000001170](https://doi.org/10.1097/JCP.0000000000001170)]
73. Metse AP, Wiggers J, Wye P, Wolfenden L, Freund M, Clancy R, et al. Efficacy of a universal smoking cessation intervention initiated in inpatient psychiatry and continued post-discharge: a randomised controlled trial. *Australian and New Zealand Journal of Psychiatry* 2017;**51**(4):366-81. [DOI: [10.1177/0004867417692424](https://doi.org/10.1177/0004867417692424)]
74. ACTRN12609000465257. A randomised controlled trial examining the efficacy of a smoking cessation intervention initiated during psychiatric hospitalisation on post-discharge smoking reduction, nicotine dependence and abstinence. <https://www.anzctr.org.au/Trial/Registration/TrialReview.aspx?id=83938> (first registered 16 June 2009).
75. ACTRN12612001042831. Evaluating the efficacy of an integrated smoking cessation intervention for mental health patients: a randomised controlled trial. <https://www.anzctr.org.au/Trial/Registration/TrialReview.aspx?id=363069&isReview=true> (first registered 28 September 2012).
76. Metse AP, Wiggers J, Wye P, Clancy R, Moore, L, Adams M, et al. Uptake of smoking cessation aids by smokers with a mental illness. *Journal of Behavioral Medicine* 2016;**39**:876-86. [DOI: [10.1007/s10865-016-9757-3](https://doi.org/10.1007/s10865-016-9757-3)]
77. Prochaska JJ, Hall SE, Delucchi K, Hall SM. Efficacy of initiating tobacco dependence treatment in inpatient psychiatry: a randomized controlled trial. *American Journal of Public Health* 2014;**104**(8):1557-65. [DOI: [10.2105/AJPH.2013.301403](https://doi.org/10.2105/AJPH.2013.301403)]
78. NCT00136812. Treating tobacco dependence in inpatient psychiatry - 1. <https://clinicaltrials.gov/study/NCT00136812> (first registered 29 August 2005).
79. Gelkopf M, Noam S, Rudinski D, Lerner A, Behrbalk P, Bleich A, et al. Nonmedication smoking reduction program for inpatients with chronic schizophrenia: a randomized control design study. *Journal of Nervous and Mental Disease* 2012;**200**(2):142-6. [DOI: [10.1097/NMD.0b013e3182438e92](https://doi.org/10.1097/NMD.0b013e3182438e92)]
80. Hickman NJ, Delucchi KL, Prochaska JJ. Treating tobacco dependence at the intersection of diversity, poverty, and mental illness: a randomized feasibility and replication trial. *Nicotine & Tobacco Research : Official Journal of the Society for Research on Nicotine and Tobacco* 2015;**17**(8):1012-21. [DOI: [10.1093/ntr/ntv034](https://doi.org/10.1093/ntr/ntv034)]
81. NCT00136812. Treating tobacco dependence in inpatient psychiatry - 1. <https://clinicaltrials.gov/study/NCT00136812> (first posted 29 August 2005).
82. Tavakoli-Ardakani M, Gholamzadeh SZ, Beyraghi N, Najarimoghadam S, Kheradmand A. Comparison between cytosine and nicotine replacement therapy in smoking cessation among inpatient psychiatric patients. *Journal of Addictive Diseases* 2023;**42**(4):1-8. [DOI: <https://dx.doi.org/10.1080/10550887.2023.2237395>]
83. IRCT20100127003210N16. Cytisine versus nicotine for smoking cessation in hospitalized psychiatric patients. <https://trialsearch.who.int/Trial2.aspx?TrialID=IRCT20100127003210N16> (first registered 14 January 2019).
84. Chen H-K, Lan T-H, Wu B-J. A double-blind randomized clinical trial of different doses of transdermal nicotine patch for smoking reduction and cessation in long-term hospitalized schizophrenic patients. *European Archives of Psychiatry and Clinical Neuroscience* 2013;**263**(1):75-82. [DOI: [10.1007/s00406-012-0338-3](https://doi.org/10.1007/s00406-012-0338-3)]
85. Stockings E, Bowman JA, Baker AL, Terry M, Clancy R, Wye PM, et al. Impact of a postdischarge smoking cessation intervention for smokers admitted to an inpatient psychiatric facility: a randomized controlled trial. *Nicotine & Tobacco Research : Official Journal of the Society for Research on Nicotine and Tobacco* 2014;**16**(11):1417-28. [DOI: [10.1093/ntr/ntu097](https://doi.org/10.1093/ntr/ntu097)]
86. Stockings E, Bowman JA, Wiggers J, Baker A, Terry M, Clancy R, et al. A randomised controlled trial linking mental health inpatients to community smoking cessation supports: a study protocol. *BioMed Central Public Health* 2011;**11**(1):article 570. [DOI: [10.1186/1471-2458-11-570](https://doi.org/10.1186/1471-2458-11-570)]
87. Akbarpour F, Rezaei O, Khodaie-Ardakani MR, Sheikhatvan M, Goodarzi H, Dolatshahi B. A double-blind placebo-controlled trial of bupropion for smoking abstinence and cognition improvement in schizophrenia. *Minerva Psichiatrica* 2010;**51**(4):263-9.
88. Das S, Hickman NJ, Prochaska JJ. Treating smoking in adults with co-occurring acute psychiatric and addictive Disorders. *Journal of Addiction Medicine* 2017;**11**(4):273-279. [DOI: <https://dx.doi.org/10.1097/ADM.0000000000000320>]
89. Richard A Brown. Intervening to promote tobacco cessation following psychiatric hospitalization. <https://clinicaltrials.gov/study/NCT05672914> 2022;-(-)-. [CLINICALTRIALS.GOV: <https://clinicaltrials.gov/study/NCT05672914>]
90. Bennett ME. Pilot testing a digital intervention to improve smoking cessation in persons with serious mental illness. ClinicalTrials.gov 2020. [CLINICALTRIALS.GOV: <https://clinicaltrials.gov/study/NCT04367506>]
91. McKenzie JE, Brennan SE. Chapter 12: Synthesizing and presenting findings using other methods (last updated October 2019). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li

T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.5. Cochrane, 2024. Available from <https://www.cochrane.org/handbook>.

92. Endicott J, Nee J, Harrison W, Blumenthal R. Quality of Life Enjoyment and Satisfaction Questionnaire: a new measure. *Psychopharmacology Bulletin* 1993;**29**(2):321-6. [PMID: europepmc.org/abstract/MED/8290681]

93. Kay SR, Fiszbein A, Opler LA. The Positive and Negative Syndrome Scale (PANSS) for Schizophrenia. *Schizophrenia Bulletin* 1987;**13**(2):261-76. [DOI: [10.1093/schbul/13.2.261](https://doi.org/10.1093/schbul/13.2.261)]

94. Mendez M. Chapter 2 - principles of mental status testing. In: *Mental Status Examination Handbook*. Philadelphia: PA: Elsevier Inc., 2022:9-15. [DOI: [10.1016/B978-0-323-69489-6.00002-4](https://doi.org/10.1016/B978-0-323-69489-6.00002-4)]

95. Anthenelli RM, Benowitz NL, West R, St Aubin L, McRae T, Lawrence D, et al. Neuropsychiatric safety and

efficacy of varenicline, bupropion, and nicotine patch in smokers with and without psychiatric disorders (EAGLES): a double-blind, randomised, placebo-controlled clinical trial. *Lancet* 2016;**387**(10037):2507-20. [DOI: [10.1016/S0140-6736\(16\)30272-0](https://doi.org/10.1016/S0140-6736(16)30272-0)]

96. Stead LF, Koilpillai P, Fanshawe TR, Lancaster T. Combined pharmacotherapy and behavioural interventions for smoking cessation. *Cochrane Database of Systematic Reviews* 2016, Issue 3. Art. No: CD008286. [DOI: [10.1002/14651858.CD008286.pub3](https://doi.org/10.1002/14651858.CD008286.pub3)]

97. Stead LF, Carroll AJ, Lancaster T. Group behaviour therapy programmes for smoking cessation. *Cochrane Database of Systematic Reviews* 2017, Issue 3. Art. No: CD001007. [DOI: [10.1002/14651858.CD001007.pub3](https://doi.org/10.1002/14651858.CD001007.pub3)]

98. Prochaska JJ, Das S, Young-Wolff KC. Smoking, mental illness, and public health. *Annual Review of Public Health* 2017;**38**:165-85. [PMID: [10.1146/annurev-publhealth-031816-044618](https://pubmed.ncbi.nlm.nih.gov/281146/)]

ADDITIONAL TABLES

Table 1. Overview of included studies and synthesis table

Study name (year) country	Study design	Psychiatry inpatient setting	Adult population (total: N, n = male/female)	Intervention (n)	Comparator (n)	Outcome domains	Outcome measure	Outcome time point	Method of synthesis
Akbarpour 2010 Iran	RCT	Setting not stated	Schizophrenia Total: 32 Male: 32 Female: 0	Bupropion 150 to 300 mg (n = 16)	Placebo (n = 16)	Reduction in number CPD Adverse events Psychiatric symptoms (cognitive impairment)	CPD change MMSE change	8 weeks 8 weeks	Narrative Narrative
Brown 2021 USA	RCT	Acute care private psychiatric hospital	All diagnoses Total: 342 Male: 173 Female: 169	Brief behavioural counselling plus NRT with follow-up community support (n = 169)	Usual care (n = 173)	Abstinence Uptake cessation support Serious adverse events	PPA biochemically verified Use of any cessation supports Deaths	6 months 6 months 12 months	Meta-analysis Narrative Meta-analysis
Chan 2020 Taiwan	Cluster-RCT	Acute psychiatric	All diagnoses Total: 310 Male: 241 Female: 69	NRT gum (n = 71) vs patch (n = 55) vs inhaler (184)	NRT gum (n = 71) vs patch (n = 55) vs inhaler (184)	Abstinence Adverse events psychiatric symptoms Weight	Exhaled CO2 concentration CGI-S Weight change	8 weeks 8 weeks 8 weeks	Narrative Narrative Narrative
Chen 2013 Taiwan	RCT	Long stay unit	Schizophrenia Total: 184 Male: 171 Female: 13	NRT patch High dose (n = 92)	NRT patch Low dose (n = 92)	Abstinence Reduction in number CPD Adverse events Psychiatric symptoms	PPA biochemically verified CPD change PANSS	8 weeks 8 weeks 8 weeks	Narrative Narrative Narrative
Gelkopf 2012 Israel	RCT	Long stay unit	Schizophrenia-type Total: 53 Male: 37 Female: 16	Group behavioural smoking reduction (n = 35)	Waitlist (n = 35)	Reduction in number CPD Adverse events Psychiatric symptoms Weight	CPD change PANSS and CGI-S Weight Change Q-LES-Q-18	3 months 3 months 3 months	Narrative Narrative Narrative

Table 1. Overview of included studies and synthesis table *(Continued)*

						Quality of life			
Hickman 2015	RCT	Non-acute & acute units	All diagnoses Total: 100 Male: 65 Female: 32 Transgender: 3	Brief behavioural counselling plus NRT with follow-up community support (n = 49)	Usual care (n = 51)	Abstinence Serious adverse events	PPA biochemical-ly verified Deaths Psychiatric rehospitalisation	3 and 6 months 12months	Meta-analy-sis Meta-analy-sis Narrative
Metse 2017	RCT	Public in-patient	All diagnoses Total: 754 Male: 462 Female: 292	Brief behavioural counselling plus NRT with follow-up community support (n = 373)	Usual care (n = 367)	Abstinence Uptake cessation support ^a Smoking reduction Serious adverse events	PPA biochemical-ly verified Use of interven-tion supports CPD change Deaths	6 months 1 month 6 months 12months	Meta-analy-sis Narrative Narrative Meta-analy-sis
Prochaska 2014	RCT	Inpatient psychiatry	All diagnoses Total: 224 Male:134 Female: 84 Transgender: 6	Brief behavioural counselling plus NRT with follow-up community support (n = 113)	Usual care (n = 111)	Abstinence Serious adverse events	PPA biochemical-ly verified Deaths Psychiatric hospi-talisation	3 and 6 months 18 months	Meta-analy-sis Narrative Meta-analy-sis Narrative
Stockings 2014	RCT	Acute & comor-bid psychi-atric sub-stance use	All diagnoses Total: 205 Male: 110 Female: 95	Brief behavioural counselling plus NRT with follow-up community support (n = 104)	Usual care (n = 101)	Abstinence Uptake cessation sup-port Smoking reduction Adverse events Psychiatric symptoms	PPA biochemical-ly verified Use of interven-tion supports CPD reduction K10	4 and 6 months 1 month 6 months 1 month	Meta-analy-sis Narrative Narrative Narrative
Tavakoli-Ardakani 2024	RCT	Psycho-somatic ward	All diagnoses Total: 47 Male: 41	Cytisine (n = 17)	NRT gum (n = 30)	Abstinence Smoking reduction	Self-reported ab-stinence CPD reduction	6 months 8 weeks	Narrative Narrative

Table 1. Overview of included studies and synthesis table *(Continued)*

Female: 6

CGI-S: Clinical Global Impression–Severity; CPD: cigarettes smoked per day; K10: Kessler Psychological Distress Scale; MMSE: Mini-Mental State Examination; N: number; NRT: nicotine replacement therapy; PPA: point-prevalence abstinence; PANSS: Positive and Negative Syndrome Scale; Q-LES-Q-18: Quality of Life Enjoyment and Satisfaction Questionnaire - abbreviated version; RCT: randomised controlled trial; vs: versus

^aThis outcome was reported in a companion paper [76].

Table 2. Change in the number of cigarettes per day (CPD) - summary of study results

Study (year)	Comparison	Outcome time point	Reported data - mean (SD) or events/total	Study-reported effect direction
Metse 2017	Counselling + NRT & post-discharge support vs usual care	6 months	Intervention 9.6 (20.7) ^a vs usual care 12 (20.5) ^a	Favours intervention Intervention may reduce smoking by 2.4 CPD more than usual care.
Stockings 2014	Counselling + NRT & post-discharge follow-up vs usual care	6 months	Intervention 11.9 (1.1) vs usual care 19 (1.5)	Favours intervention Intervention may reduce smoking by 7.1 CPD more than usual care.
Gelkopf 2012	Group behavioural smoking reduction vs waitlist	3 months	Group behavioural programme 14.6 (9.9) vs waitlist 17.39 (5.4)	Favours intervention Intervention reduced smoking by 2.79 CPD more than waitlist.
Akbarpour 2010	Pharmacotherapy (medications) Bupropion 150 to 300mg vs placebo	8 weeks	Bupropion 11.1 (8.8) vs placebo 13.4 (11.8)	Favours intervention Bupropion reduced smoking by 2.3 CPD more than placebo.
Tavakoli 2024	Pharmacotherapy (medications) Cytisine vs NRT (gum)	8 weeks	Cytisine 5.2 (3.4) vs NRT 10.4 (3)	Favours intervention Cytisine reduced smoking by 5.2 CPD more than NRT gum.
Chen 2013	Pharmacotherapy (NRT forms/doses) NRT: nicotine transdermal patch High dose vs low dose	8 weeks	High-dose NRT 12 (9.2) vs low-dose NRT 10.1 (7.7)	Favours low-dose NRT Low-dose patch reduced smoking by 1.9 CPD more than high-dose patch.

CPD: cigarettes smoked per day; NRT: nicotine replacement therapy; SD: standard deviation; vs: versus

^aSD calculated from reported SE using $SD = SE \times \sqrt{n}$

Table 3. Non-pharmacological adverse events - summary of study results

Outcome	Study (year)	Outcome measure(s)	Outcome time point	Reported data - mean (SD) or events/total	Study-reported effect direction
Psychiatric symptoms	Stockings 2014	Kessler Psychological Distress Scale – 10 (K10) Higher scores = greater psychological distress	6 months	K10 score Intervention: 21.3 (1.1) Usual care: 22.0 (1.1) MD (–0.7, 95% CI –3.7 to 2.3) P = 0.642	No difference
	Gelkopf 2012	Positive and negative syndrome scale (PANSS) Higher scores = greater symptom severity Hamilton Depression Rating Scale (HAM-D) Higher scores = greater depression severity	3 months	PANSS score Intervention baseline 63.87 (21.1) to follow-up 55.41 (21.1) Waitlist baseline 78.25 (26.5) to follow-up 79.58 (26.2) ^a Mean change: Intervention –8.46 vs Waitlist +1.33 = –9.79 points Significant time x group x PANSS change interaction reported (F = 3.15, P = 0.05) HAM-D = no change reported in depression	Favours intervention Intervention may improve psychiatric symptoms compared to waitlist.
	Chen 2013	Positive and negative syndrome scale (PANSS) Higher scores = greater symptom severity	8 weeks	Mean change High dose: –2.5 (11) vs Low dose: –2.5 (11)	No difference
	Chan 2020	Clinical Global Impressions – Severity (CGI-S) Higher scores = greater symptom severity	8 weeks	Mean change Inhaler: –1.0 (1.2) (P < 0.001) vs Gum: 0.7 (1.1) (P < 0.001) vs Patch: –0.3 (0.8) (P = 0.002) Inhaler largest decrease (F = 17.019, P < 0.001)	Favours inhaler

Table 3. Non-pharmacological adverse events - summary of study results *(Continued)*

	Akbarpour 2010	Mini-Mental State Examination (MMSE) Higher scores = better global cognition	8 weeks	Bupropion baseline 23.5 (4.4) to follow-up 26.3 (3.5) $P = 0.002$ vs Placebo baseline 22.8 (4.6) to follow-up 22.8 (3.5) $P = 0.954$ Mean change: Bupropion +3.5 points vs Placebo 0 points	Favours intervention Bupropion may reduce cognitive impairment by 3.5 points more than placebo.
Weight change	Gelkopf 2012	Body weight (kg)	3 months post-intervention	Intervention baseline 75.53 (15.1) to follow-up 71.71 (18.9) No significant difference between groups ($P = 0.20$)	No difference
	Chan 2020	Body weight (kg)	8 weeks	Mean change Inhaler: 1.5 (3.2) ($P < 0.001$) vs Gum: 0.8 (5.6) ($P = 0.212$) vs Patch: 0.2 (3.2) $P = 0.717$ No significant difference between groups ($P = 0.080$)	No difference

K10: Kessler Psychological Distress Scale; CGI-S: Clinical Global Impression–Severity; HAM-D: Hamilton Depression Rating Scale; MMSE: Mini-Mental State Examination; PANSS: Positive and Negative Syndrome Scale; P: probability; SD: standard deviation; vs: versus